OLS, Ridge, and Lasso Through One Worked Example

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Source: `Lectures/L08_student_ols_ridge_lasso_worked_example.pdf` (54 slides, with some in-class annotations) · Previous: Ridge, lasso and elastic net · Next: Metrics, thresholds and costs

> **How to use this file.** Lecture 7 told the story *correlated predictors → flat valley → small eigenvalue → unstable OLS → ridge/lasso fix*. Lecture 8 **computes every object in that story** for one 4-row dataset and draws it. Part A collects the small amount of new maths (2×2 inverses, crossing lines, gradient-descent rates, the lasso optimality conditions). Part B goes through all 54 slides with every number recomputed. Part C is practice.

The whole lecture in one table (sample A vs a slightly perturbed sample B):

	sample A	sample B	jump between samples
OLS	$(0.8,\ 1.2)$	$(1.2,\ 0.8)$	**0.566**
Ridge, $\lambda = 1$	$(0.9713,\ 0.9801)$	$(0.9790,\ 0.9721)$	**0.011**
Lasso, $\lambda = 20$	$(0,\ 1.0004)$	$(0,\ 0.7994)$	one twin dropped

The responses moved by 0.08. OLS coefficients moved 7× that; ridge moved 1/7 of it. **Same data, same noise; the difference is the shape of the surface being minimised.**

Step-up path

You need. The ridge and lasso formulas from the previous note.

1. Solve a small system by hand, and see two equations as two crossing lines.
2. Make the columns nearly equal, and watch the crossing point run away.
3. Rotate into eigen-directions, where gradient descent splits into separate problems.
4. Compute the lasso solution from its optimality condition.

You can stop when you can predict from eigenvalues how many gradient steps a problem will need.

Contents

Part A — Maths from the ground up

1. The running dataset and its two objects $X^\top X$, $X^\top y$
2. A 2×2 system = two crossing lines
3. Eigen-directions of a near-duplicate Gram matrix
4. The loss around its minimum: a valley
5. Gradient descent: the $1 - \eta\mu$ factor
6. Lasso optimality: when is a coefficient exactly zero?
7. Train / validation / test

Part B — Slides · All 54 slides

Part C — Practice · Jump

Part A Maths From the Ground Up

A1 The running dataset and its two objects

i	x_{i1}	x_{i2}	$y_i^{(A)}$	$y_i^{(B)}$
1	−3	−3.1	−6.12	−6.08
2	−1	−0.9	−1.88	−1.92
3	1	1.1	2.12	2.08
4	3	2.9	5.88	5.92

Both columns are centred (sum 0), so no intercept is needed. Model: $\hat{y} = \omega_1 x_1 + \omega_2 x_2$ (the slides use ω for coefficients).

Gram matrix (dot products of columns, L5 A3):

- $x_1^\top x_1 = 9 + 1 + 1 + 9 = 20$;
- $x_1^\top x_2 = 9.3 + 0.9 + 1.1 + 8.7 = 20$;
- $x_2^\top x_2 = 9.61 + 0.81 + 1.21 + 8.41 = 20.04$.

$$X^\top X = \begin{bmatrix} 20 & 20 \\ 20 & 20.04 \end{bmatrix}, \qquad \det = 20(20.04) - 20^2 = 0.8.$$

Feature–response alignment:

- $x_1^\top y^{(A)} = 18.36 + 1.88 + 2.12 + 17.64 = 40$;
- $x_2^\top y^{(A)} = 18.972 + 1.692 + 2.332 + 17.052 = 40.048$.

$X^\top y^{(A)} = (40,\ 40.048)$, and similarly $X^\top y^{(B)} = (40,\ 40.032)$.

Correlation of the columns: $\frac{20}{\sqrt{20 \cdot 20.04}} = 0.999$. Practically the same sensor twice.

A2 A 2×2 system = two crossing lines

Inverse of a 2×2 (L4 §A7): $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Swap the diagonal, negate the off-diagonal, divide by the determinant.

The normal equation $X^\top X\omega = X^\top y$ is two equations = **two lines** in the (ω_1, ω_2) plane; the solution is where they cross.

$$20\omega_1 + 20\omega_2 = 40, \qquad 20\omega_1 + 20.04\omega_2 = 40.048.$$

Slopes: $-20/20 = -1.000$ and $-20/20.04 = -0.998$. **Nearly parallel lines cross at a point that moves a lot when either line shifts a little.** That's ill-conditioning in a picture: a small determinant (0.8, tiny next to entries of size 20) ⇒ large entries in the inverse ⇒ big amplification.

Subtract the equations to see it: $0.04\omega_2 = 0.048 \Rightarrow \omega_2 = 1.2$, then $\omega_1 = 2 - 1.2 = 0.8$. For sample B: $0.04\omega_2 = 0.032 \Rightarrow \omega_2 = 0.8$. A change of 0.016 in the right-hand side, divided by 0.04, moved ω_2 by 0.4.

A3 Eigen-directions of a near-duplicate Gram matrix

For $\begin{bmatrix} a & b \\ b & d \end{bmatrix}$: $\mu = \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}$ with $T = 40.04$, $D = 0.8$:

$$\mu = 20.02 \pm \sqrt{400.8004 - 0.8} = 20.02 \pm 20.0000 \;\Rightarrow\; \mu_{\text{strong}} = 40.02, \quad \mu_{\text{weak}} = 0.01999.$$

Check: sum $= 40.04 = T$ ✓, product $\approx 0.8 = D$ ✓.

	direction q	μ	$\|Xq\| = \sqrt{\mu}$	meaning
strong	$(0.7068,\ 0.7075) \approx \frac{1}{\sqrt{2}}(1, 1)$	40.02	6.326	both coefficients up together: predictions move a lot
weak	$(-0.7075,\ 0.7068) \approx \frac{1}{\sqrt{2}}(-1, 1)$	0.01999	0.141	one up, one down: predictions barely move

Condition number $\kappa = 40.02/0.01999 = 2002$.

A4 The loss around its minimum: a valley

Because J is quadratic and $\nabla J(\hat{\omega}) = 0$ (L5 slide 23, L7 slide 9):

$$J(\hat{\omega} + tq_j) = J(\hat{\omega}) + \tfrac{1}{2}\mu_j t^2.$$

step t along	weak ($\frac{1}{2}\mu = 0.01$)	strong ($\frac{1}{2}\mu = 20.01$)
0.25	0.0006	1.25
0.5	0.0025	5.0
1	0.01	20.0

Contours of J are ellipses with axes along q_j and half-lengths $\propto 1/\sqrt{\mu_j}$: ratio $\sqrt{2002} = 44.7$. **A valley 45 times longer than it is wide.**

A5 Gradient descent: the $1 - \eta\mu$ factor

GD on a quadratic: $\omega_{t+1} = \omega_t - \eta\nabla J = \omega_t - \eta G(\omega_t - \hat{\omega})$. Let $e_t = \omega_t - \hat{\omega}$ be the error. Then

$$e_{t+1} = (I - \eta G)e_t.$$

In eigen-coordinates each component is multiplied by $(1 - \eta\mu_j)$ **every step**:

$\eta = 0.02$	strong ($\mu = 40.02$)	weak OLS ($\mu = 0.02$)	weak ridge ($\mu + \lambda = 1.02$)
factor $1 - \eta\mu$	0.1996	0.9996	0.9796
after 400 steps	≈ 0	$0.9996^{400} = 0.85$	$0.9796^{400} = 2.6 \times 10^{-4}$

The step size must be small enough for the **steep** direction ($\eta < 2/\mu_{\max} = 0.05$), so progress along the **flat** direction is a crawl. Ridge lifts the flat eigenvalue from 0.02 to 1.02 and turns the crawl into a walk.

A6 Lasso optimality: when is a coefficient exactly zero?

Objective $J(\omega) = \frac{1}{2}\|y - X\omega\|^2 + \lambda\sum_j|\omega_j|$. Generalising L7 A7's 1-D rule, the optimum satisfies, for each j:

- if $\omega_j \neq 0$: $x_j^\top (y - X\omega) = \lambda \operatorname{sign}(\omega_j)$ (data pull = friction);
- if $\omega_j = 0$: $|x_j^\top (y - X\omega)| \leq \lambda$ (pull too weak to overcome friction).

Exact lasso path for sample A.

Phase 1, both nonzero and positive. Stack the two equations: $X^\top X\omega = X^\top y - \lambda(1,1) \Rightarrow$

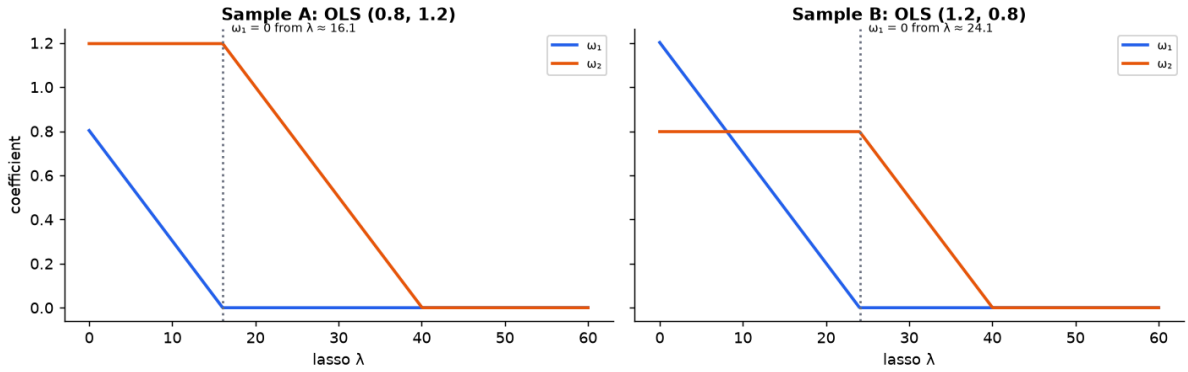
$$\omega = \hat{\omega}_{\text{OLS}} - \lambda(X^\top X)^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 1.2 \end{bmatrix} - \lambda \cdot \frac{1}{0.8} \begin{bmatrix} 20.04 - 20 \\ -20 + 20 \end{bmatrix} = \begin{bmatrix} 0.8 - 0.05\lambda \\ 1.2 \end{bmatrix}.$$

ω_2 stays at 1.2 while ω_1 slides down and hits 0 at $\lambda = 0.8/0.05 = 16$.

Phase 2, only ω_2 active. $x_2^\top (y - x_2\omega_2) = \lambda \Rightarrow \omega_2 = \frac{40.048 - \lambda}{20.04}$. At $\lambda = 20$: $\omega_2 = 20.048/20.04 = 1.0004$ ✓ (slide 46). Hits 0 at $\lambda = 40.048$.

Check ω_1 really stays 0: $|x_1^\top (y - x_2\omega_2)| = 40 - 20 \cdot \frac{40.048 - \lambda}{20.04} = 0.032 + 0.998\lambda \leq \lambda \iff \lambda \geq 16$ ✓.

Sample B: $\omega = (1.2 - 0.05\lambda,\ 0.8)$; $\omega_1 = 0$ from $\lambda = 24$; then $\omega_2 = (40.032 - \lambda)/20.04$. In **both** samples lasso keeps x_2, because its column is 0.2% longer and its alignment with y is larger (slide 47).



Lasso paths for both samples, computed exactly: ω_1 slides to zero ($\lambda = 16$ for A, 24 for B) while ω_2 holds, then ω_2 shrinks.

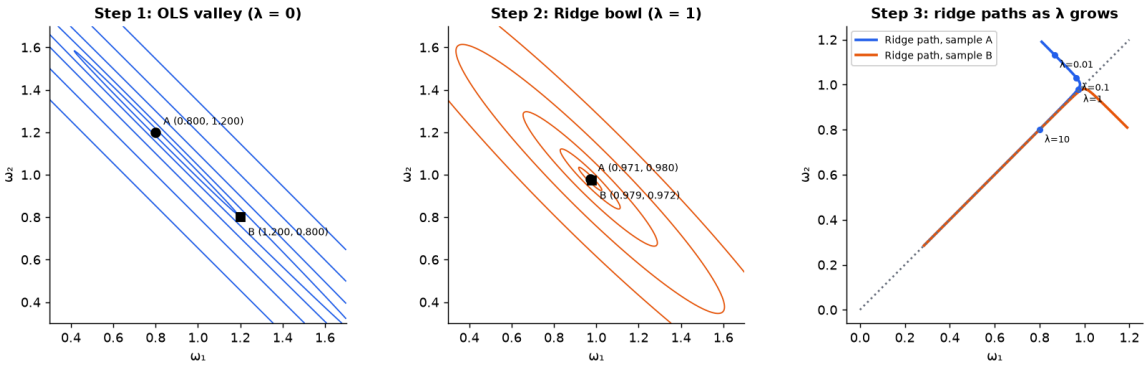
(The slide quotes 16.8 and 25.1, read off its plotted λ grid; the exact kinks from this algebra are 16 and 24.)

A7 Train / validation / test

set	used for	touched how often
training	fit ω for every candidate λ; compute standardisation m_j, s_j	many times
validation	choose λ^* (and the method)	once per candidate
test	report the final error	**exactly once**

Validation error becomes optimistic once you've chosen the λ that looked best on it. The test set must not have influenced any decision.

Part B Slide by Slide (all 54 slides)



The whole lecture in three panels: OLS valley (A and B far apart), ridge bowl (A and B on top of each other), and the ridge paths sliding onto $\omega_1 = \omega_2$ then toward 0.

Slide 1 Title

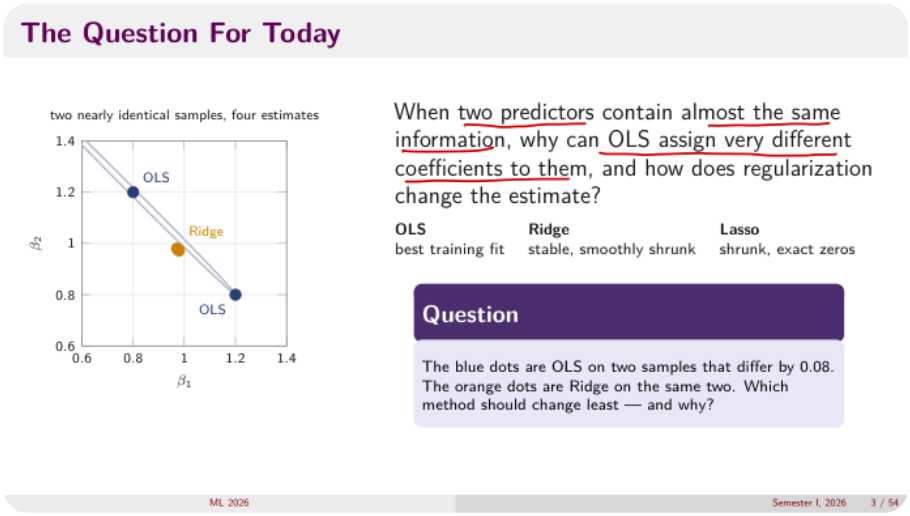
Lecture 8: OLS, Ridge, and Lasso Through One Worked Example. Prof. Udit Bhatia, Semester I 2026. (Annotation: "Assignment 1".)

Slide 2 What The Previous Lecture Established

correlated predictors → a flat valley in the loss → small eigenvalue of $X^\top X$ → unstable OLS coefficients.

> **Summary:** today computes every object in that chain for one small dataset, draws each one (the valley, its two directions, what ridge does to it), then repeats with ridge and lasso.

Slide 3 The Question For Today

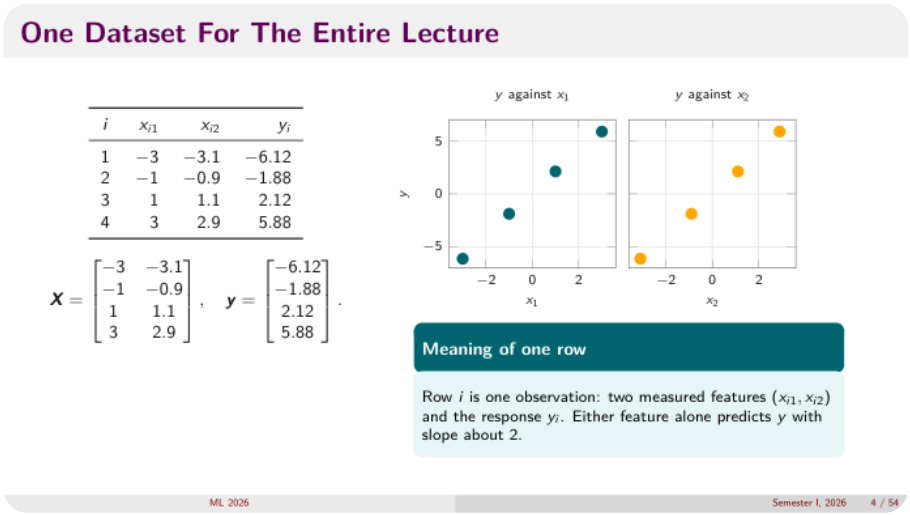


Slide 3 — Two nearly identical samples, four estimates: OLS dots far apart, ridge dots together.

When two predictors carry almost the same information, why can OLS assign very different coefficients, and how does regularisation change that? OLS = best training fit; ridge = stable, smoothly shrunk; lasso = shrunk, exact zeros.

Question: the samples differ by 0.08. Which method should change least? → Ridge (slides 30–35).

Slide 4 One Dataset For The Entire Lecture



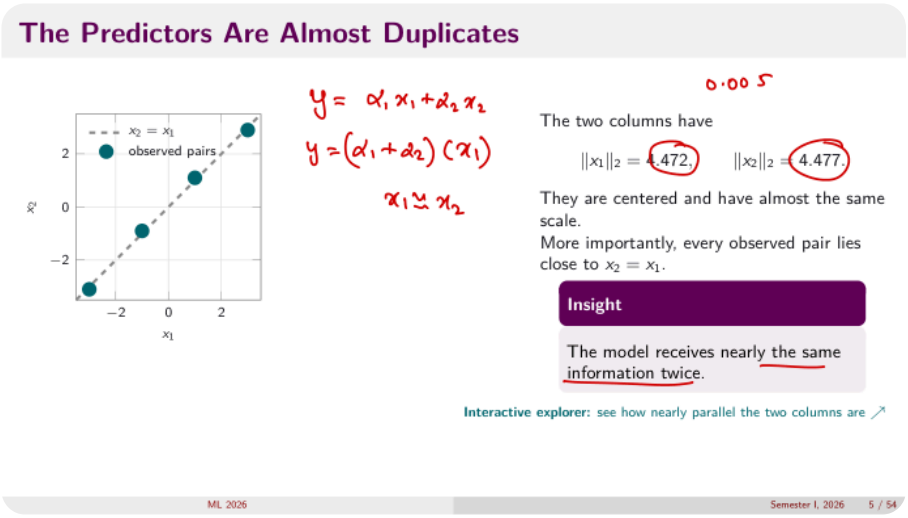
i	x_{i1}	x_{i2}	y_i
1	−3	−3.1	−6.12
2	−1	−0.9	−1.88
3	1	1.1	2.12
4	3	2.9	5.88

$$\boldsymbol{X} = \begin{bmatrix} -3 & -3.1 \\ -1 & -0.9 \\ 1 & 1.1 \\ 3 & 2.9 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} -6.12 \\ -1.88 \\ 2.12 \\ 5.88 \end{bmatrix}.$$

Slide 4 — Four rows; either feature alone predicts y with slope ≈ 2.

Table in A1. Row i = one observation: (x_{i1}, x_{i2}) and y_i. **Either feature alone predicts y with slope about 2** (e.g. $y \approx 2x_1$: $-6, -2, 2, 6$ vs observed $-6.12, -1.88, 2.12, 5.88$).

Slide 5 The Predictors Are Almost Duplicates



Slide 5 — Every (x_1, x_2) pair lies close to the line $x_2 = x_1$.

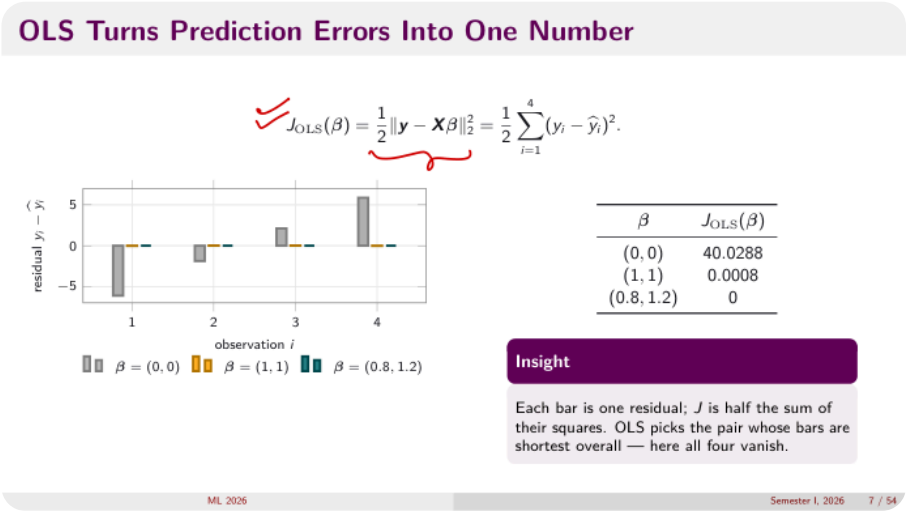
$\|x_1\| = \sqrt{20} = 4.472$, $\|x_2\| = \sqrt{20.04} = 4.477$. Centred, almost the same scale, and every pair lies close to $x_2 = x_1$ (differences only ±0.1). **The model receives nearly the same information twice.** (Annotation: $y = \omega_1 x_1 + \omega_2 x_2 \approx (\omega_1 + \omega_2) x_1$.)

Slide 6 The Prediction Rule

$\hat{y}_i = \omega_1 x_{i1} + \omega_2 x_{i2}$, $\hat{y} = X\omega$. Row 1 at $\omega = (0.8, 1.2)$: $-3(0.8) - 3.1(1.2) = -2.4 - 3.72 = -6.12$ ✓, the two stacked pieces of the first bar.

> **Summary:** features centred ⇒ intercept omitted to keep the arithmetic visible. In practice it is fitted and **left unpenalised**.

Slide 7 OLS Turns Prediction Errors Into One Number



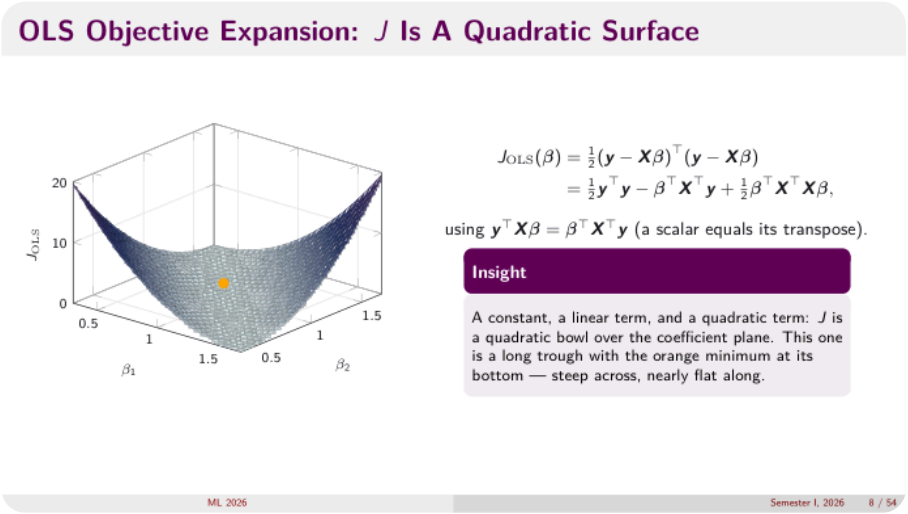
Slide 7 — Residual bars for three candidate ω.

$J_{\mathrm{OLS}}(\omega) = \frac{1}{2}\|y - X\omega\|^2 = \frac{1}{2}\sum_i (y_i - \hat{y}_i)^2.$

ω	residuals $y - X\omega$	J
$(0, 0)$	y itself	$\frac{1}{2}(37.45 + 3.53 + 4.49 + 34.57) = 40.0288$
$(1, 1)$	$y - (x_1 + x_2) = (-0.02, 0.02, 0.02, -0.02)$	$\frac{1}{2}(4 \times 0.0004) = 0.0008$
$(0.8, 1.2)$	$(0, 0, 0, 0)$	0

($(0, 0)$: $6.12^2 + 1.88^2 + 2.12^2 + 5.88^2 = 37.45 + 3.53 + 4.49 + 34.57 = 80.06$, half is 40.03.) OLS picks the pair whose bars are shortest overall; here all four vanish.

Slide 8 OLS Objective Expansion: J Is A Quadratic Surface

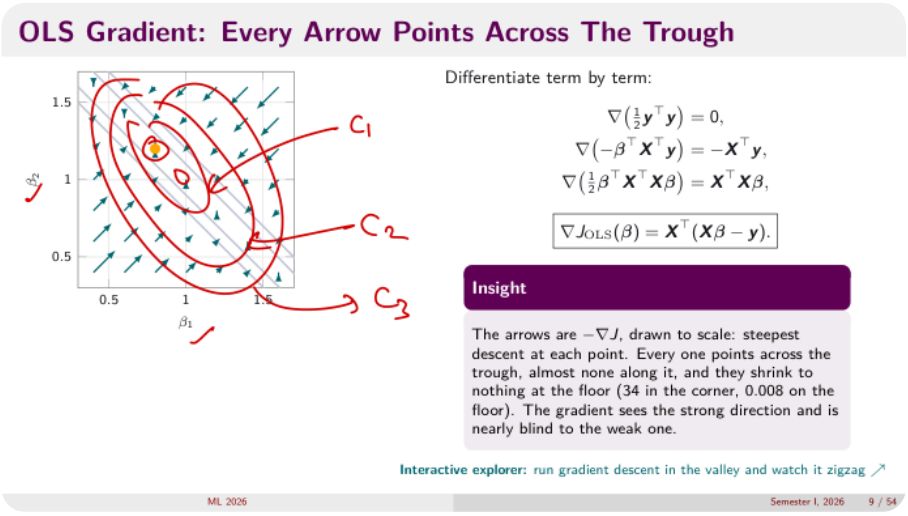


Slide 8 — A long trough with the minimum at its bottom: steep across, nearly flat along.

$$J = \tfrac{1}{2}(y - X\omega)^\top (y - X\omega) = \tfrac{1}{2} y^\top y - \omega^\top X^\top y + \tfrac{1}{2}\omega^\top X^\top X\omega$$

(using $y^\top X\omega = \omega^\top X^\top y$, a scalar equals its transpose). Constant + linear + quadratic = a **quadratic bowl** over the coefficient plane. This one is a long trough.

Slide 9 OLS Gradient: Every Arrow Points Across The Trough



Slide 9 — −∇J arrows: all point across the trough, almost none along it.

Term by term: $\nabla(\frac{1}{2} y^\top y) = 0$, $\nabla(-\omega^\top X^\top y) = -X^\top y$, $\nabla(\frac{1}{2}\omega^\top X^\top X\omega) = X^\top X\omega$. So

$$\nabla J_{\mathrm{OLS}}(\omega) = X^\top (X\omega - y).$$

Insight: the arrows are $-\nabla J$ to scale. Every one points across the trough and shrinks to nothing on the floor. **The gradient sees the strong direction and is nearly blind to the weak one.** (Annotations sketch the contour ellipses.)

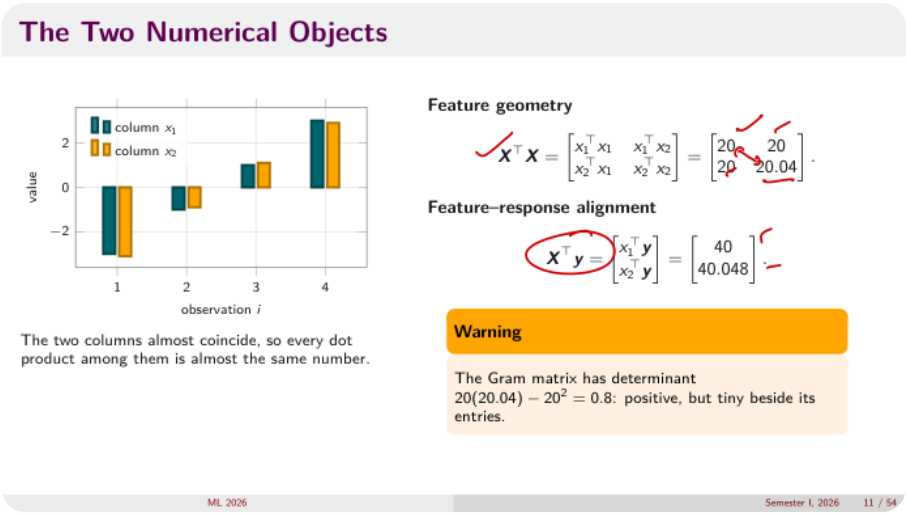
Why: $\nabla J = G(\omega - \hat{\omega})$; in eigen-coordinates its components are $\mu_j z_j$, so the weak component is multiplied by 0.02 and the strong by 40.

Slide 10 The OLS Normal Equation: Where The Arrows Vanish

$\nabla J = 0 \Rightarrow X^\top X \hat{\omega}_{\mathrm{OLS}} = X^\top y$, and if invertible $\hat{\omega} = (X^\top X)^{-1} X^\top y$. Two objects to compute: $X^\top X$ (feature geometry) and $X^\top y$ (feature–response alignment).

Insight: equally far from the minimum, $\|\nabla J\|$ is 0.008 on the floor but 11.3 off it. Reason: $\nabla J = X^\top X(\omega - \hat{\omega})$, so a displacement d along q_j gives a gradient of size $\mu_j d$: multiplied by 0.02 along the floor, by 40 across it.

Slide 11 The Two Numerical Objects

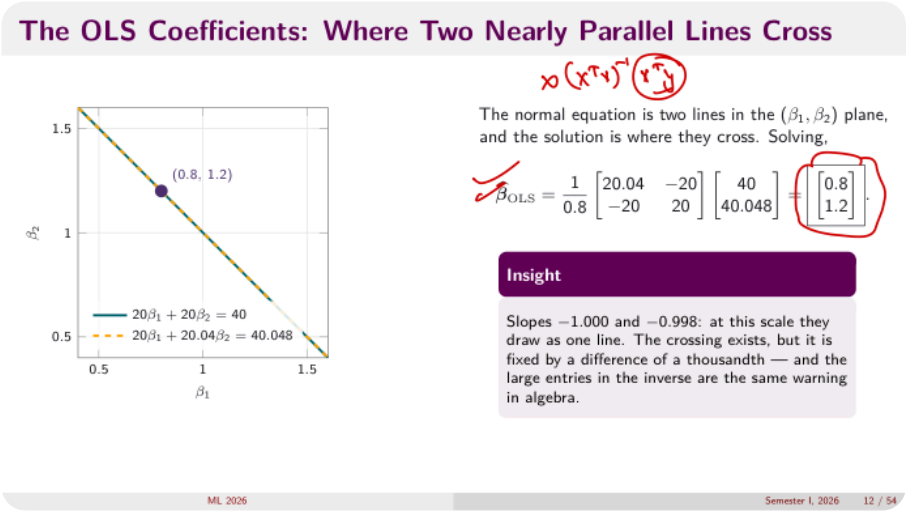


Slide 11 — The two columns almost coincide, so every dot product among them is almost the same number.

$X^\top X = \begin{bmatrix} 20 & 20 \\ 20 & 20.04 \end{bmatrix}$, $X^\top y = (40,\ 40.048)$ (all computed in A1).

Careful. $\det X^\top X = 0.8$: positive, but tiny beside its entries.

Slide 12 The OLS Coefficients: Where Two Nearly Parallel Lines Cross



Slide 12 — $20\omega_1 + 20\omega_2 = 40$ and $20\omega_1 + 20.04\omega_2 = 40.048$ cross at (0.8, 1.2).

$$\hat{\omega}_{\text{OLS}} = \frac{1}{0.8}\begin{bmatrix}20.04 & -20\\ -20 & 20\end{bmatrix}\begin{bmatrix}40\\ 40.048\end{bmatrix} = \frac{1}{0.8}\begin{bmatrix}801.6 - 800.96\\ -800 + 800.96\end{bmatrix} = \frac{1}{0.8}\begin{bmatrix}0.64\\ 0.96\end{bmatrix} = \begin{bmatrix}0.8\\ 1.2\end{bmatrix}.$$

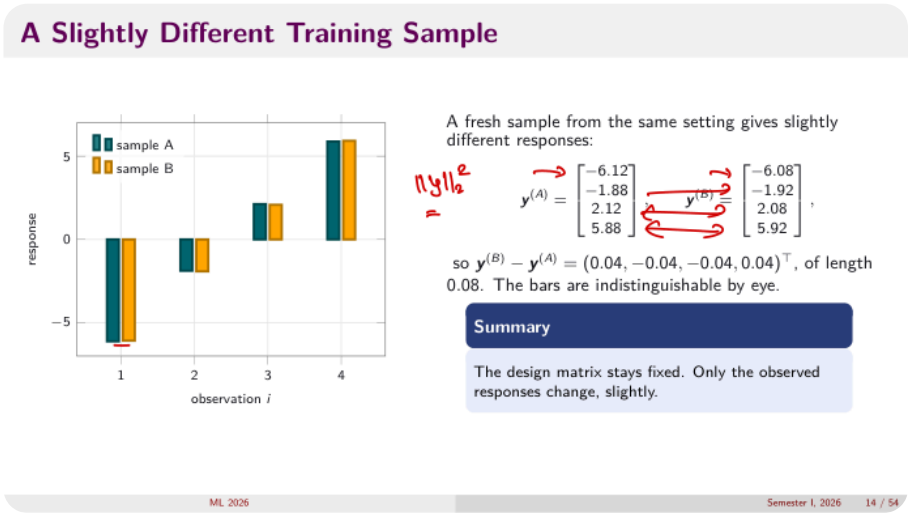
Slopes -1.000 and -0.998: at this scale they draw as one line. The crossing exists but is fixed by a difference of a thousandth; the large entries of the inverse (25.05, 25) are the same warning in algebra (A2).

Slide 13 Checking The OLS Fit

$$X\hat{\omega} = \begin{bmatrix}-3(0.8) - 3.1(1.2)\\ -0.8 - 1.08\\ 0.8 + 1.32\\ 2.4 + 3.48\end{bmatrix} = \begin{bmatrix}-6.12\\ -1.88\\ 2.12\\ 5.88\end{bmatrix} = y.$$

Zero residuals, $J = 0$. **Question:** if OLS fits perfectly, what could still be wrong? → The coefficients themselves are unreliable (slides 14–18).

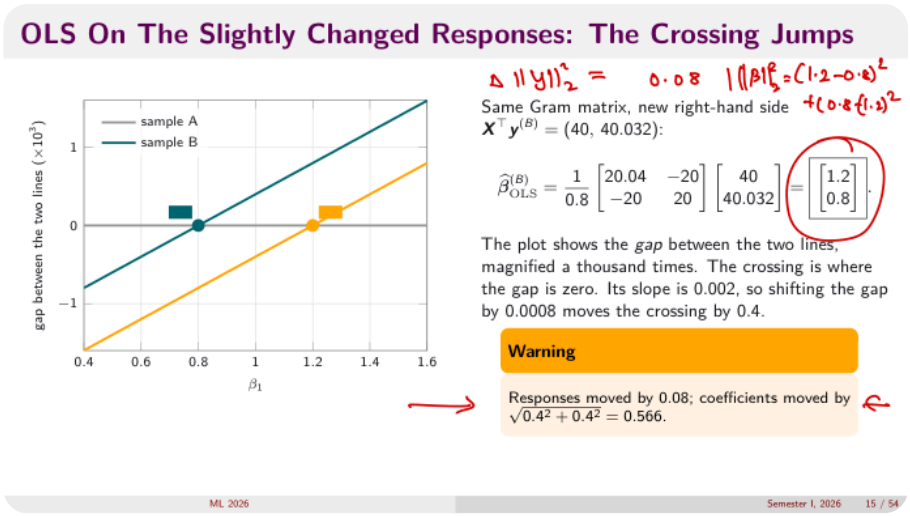
Slide 14 A Slightly Different Training Sample



Slide 14 — Samples A and B: the bars are indistinguishable by eye.

$y^{(B)} - y^{(A)} = (0.04, -0.04, -0.04, 0.04)$, length $\sqrt{4 \times 0.0016} = 0.08$. Design matrix fixed; only responses change slightly.

Slide 15 OLS On The Slightly Changed Responses: The Crossing Jumps



Slide 15 — Gap between the two lines (×1000); the crossing moves from 1.2 to 0.8.

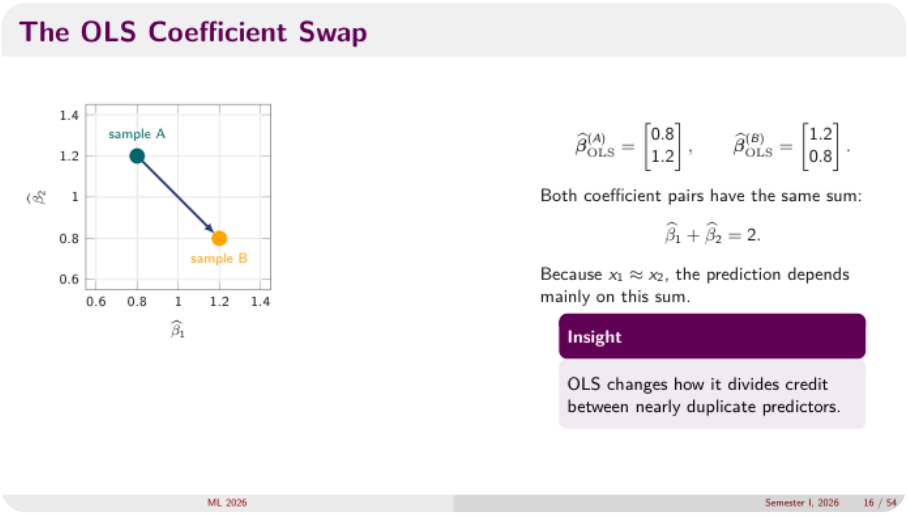
Same Gram matrix, new $X^\top y^{(B)} = (40, 40.032)$:

$$\hat{\omega}^{(B)} = \frac{1}{0.8}\begin{bmatrix} 20.04(40) - 20(40.032) \\ -20(40) + 20(40.032) \end{bmatrix} = \frac{1}{0.8}\begin{bmatrix} 0.96 \\ 0.64 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}.$$

The gap between the lines has slope 0.002, so shifting it by 0.0008 moves the crossing by 0.4.

Careful. Responses moved by 0.08; coefficients moved by $\sqrt{0.4^2 + 0.4^2} = 0.566$. (Annotation: $\|\Delta y\| = 0.08$, $\|\Delta \omega\| = 0.566$, a factor of 7.)

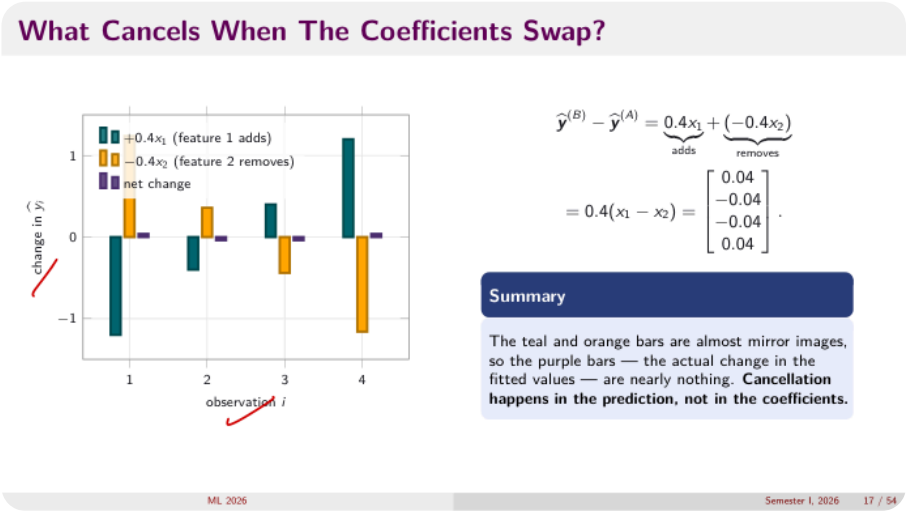
Slide 16 The OLS Coefficient Swap



Slide 16 — (0.8, 1.2) → (1.2, 0.8): a mirror swap, same sum.

Both pairs have $\hat{\omega}_1 + \hat{\omega}_2 = 2$. Since $x_1 \approx x_2$, the prediction depends mainly on this sum. **OLS changes how it divides credit between near-duplicates.**

Slide 17 What Cancels When The Coefficients Swap?

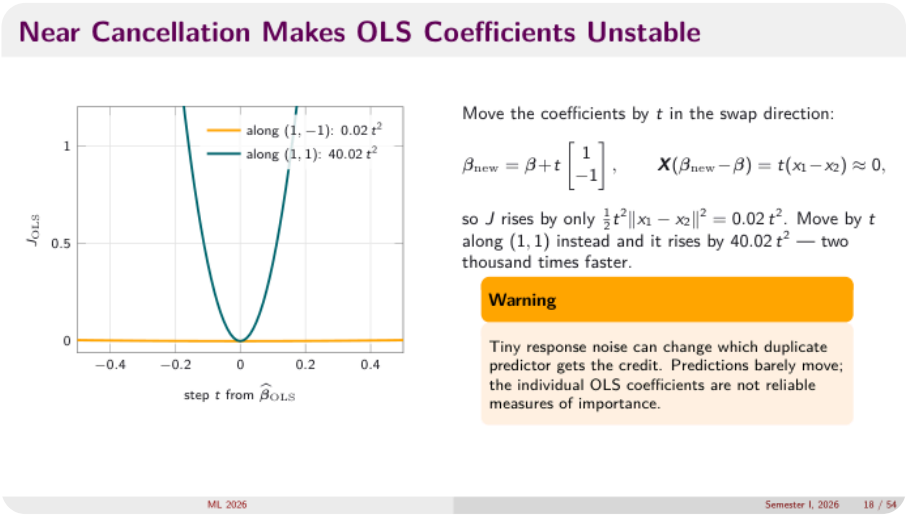


Slide 17 — $+0.4x_1$ and $-0.4x_2$ nearly cancel; the net change in $\hat{y}$ is tiny.

$$\hat{y}^{(B)} - \hat{y}^{(A)} = 0.4x_1 + (-0.4x_2) = 0.4(x_1 - x_2) = 0.4(0.1, -0.1, -0.1, 0.1) = (0.04, -0.04, -0.04, 0.04$$

Exactly $y^{(B)} - y^{(A)}$ (both fits are perfect). **Cancellation happens in the prediction, not in the coefficients.**

Slide 18 Near Cancellation Makes OLS Coefficients Unstable



Slide 18 — Along (1, −1) the loss rises as $0.02t^2$; along (1, 1) as $40.02t^2$.

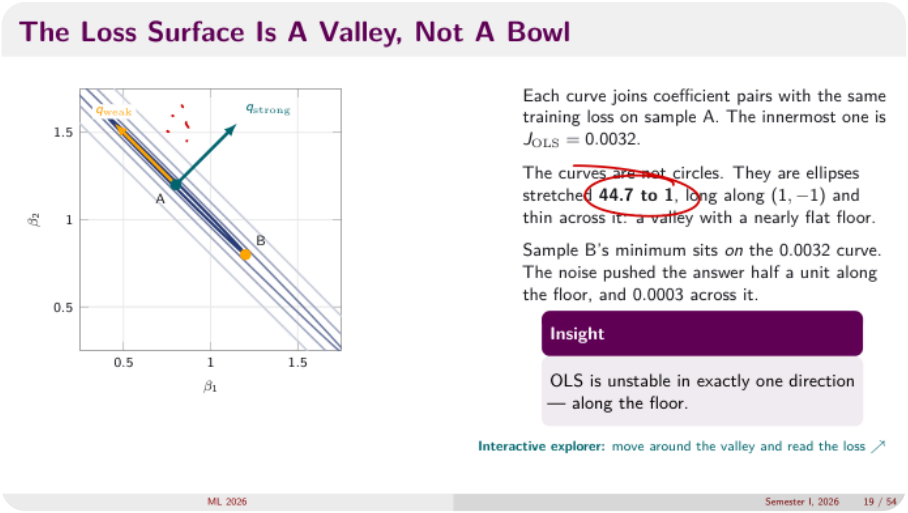
Move by $t(1,-1)$: $X(\omega_{\text{new}}-\omega)=t(x_1-x_2)$, so J rises by $\frac{1}{2}t^2\|x_1-x_2\|^2=\frac{1}{2}t^2(0.04)=0.02t^2$. Along $(1,1)$: $\frac{1}{2}t^2\|x_1+x_2\|^2=\frac{1}{2}t^2(37.21+3.61+4.41+34.81)=40.02t^2$, **two thousand times faster**.

Careful. Tiny response noise can change which duplicate gets the credit. Individual OLS coefficients are **not** reliable measures of importance.

(These use unnormalised directions; with unit vectors divide by 2 and get $\frac{1}{2}\mu t^2$ with $\mu=0.02, 40.02$.)

Slide 19

The Loss Surface Is A Valley, Not A Bowl



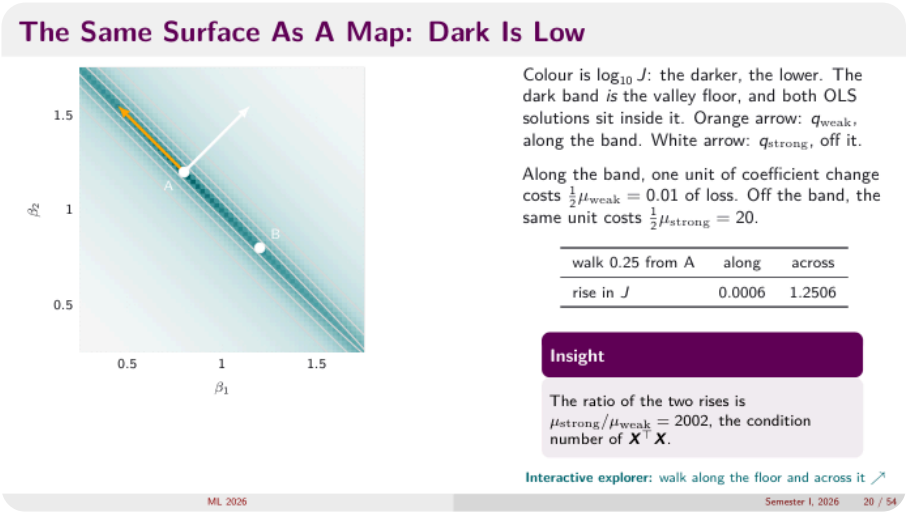
Slide 19 — Contours stretched 44.7 : 1; sample B's minimum sits on A's innermost curve, half a unit along the floor.

Each curve joins pairs with the same training loss on sample A; innermost $J=0.0032$. Ellipses stretched $\sqrt{2002}=44.7$ to 1, long along $(1,-1)$. Sample B's minimum is on that 0.0032 curve: **noise pushed it 0.57 along the floor and 0.0003 across**. Check: $J_A(1.2,0.8)=\frac{1}{2}\|y^{(A)}-y^{(B)}\|^2=\frac{1}{2}(0.0064)=0.0032$ ✓.

OLS is unstable in exactly one direction: along the floor.

Slide 20

The Same Surface As A Map: Dark Is Low



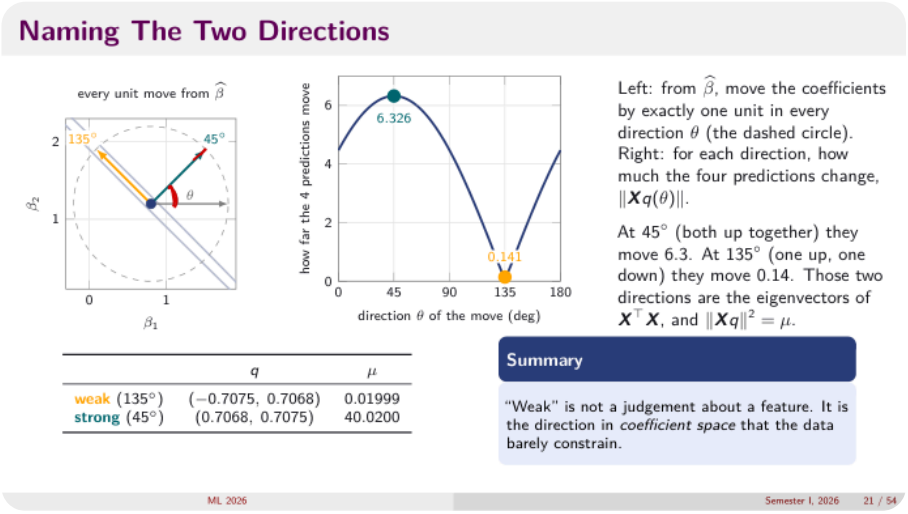
Slide 20 — Colour = $\log_{10}$ J: a dark band with both OLS solutions inside.

Along the band, one unit costs $\frac{1}{2}\mu_{\text{weak}} = 0.01$; across it, $\frac{1}{2}\mu_{\text{strong}} = 20$.

walk 0.25 from A	along	across
rise in J	$0.01(0.25)^2 = 0.0006$	$20.01(0.25)^2 = 1.2506$

Ratio $= \mu_{\text{strong}}/\mu_{\text{weak}} = 2002$, the **condition number** of $X^\top X$.

Slide 21 Naming The Two Directions

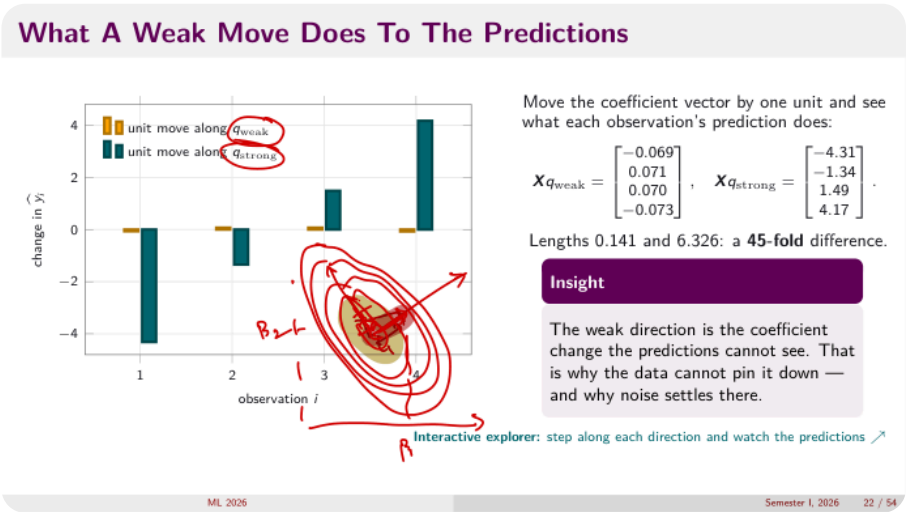


Slide 21 — Move one unit in every direction ε; predictions move 6.326 at 45° and 0.141 at 135°.

From $\hat{\omega}$, move exactly one unit in direction ε (dashed circle) and measure $\|Xq(\varepsilon)\|$, how far the four predictions move. Max at 45° (both up): 6.326. Min at 135° (one up, one down): 0.141. Those two are the **eigenvectors of** $X^\top X$, and $\|Xq\|^2 = \mu$ (A3 table).

"Weak" is not a judgement about a feature. It's the direction in coefficient space that the data barely constrain.

Slide 22 What A Weak Move Does To The Predictions

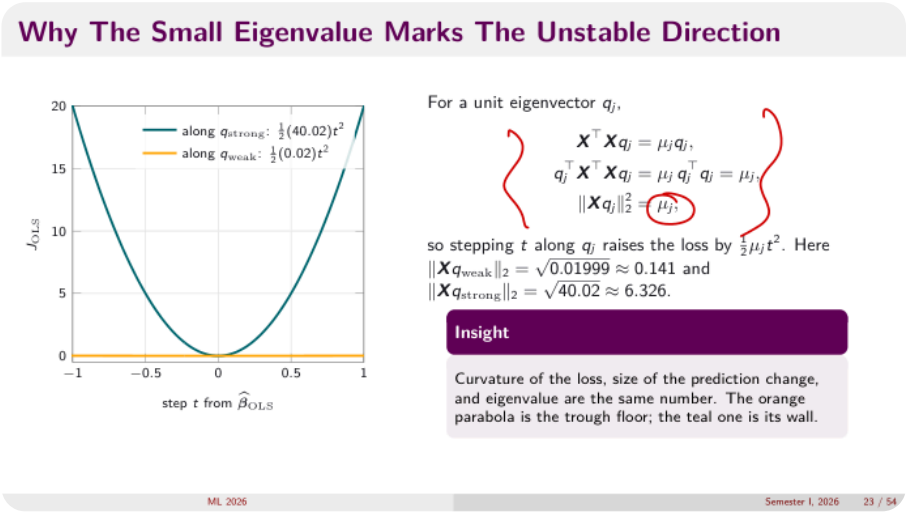


Slide 22 — Unit moves along q_weak and q_strong: prediction changes of size 0.07 vs 4.

$$Xq_{\text{weak}} = (-0.069,\ 0.071,\ 0.070,\ -0.073), \qquad Xq_{\text{strong}} = (-4.31,\ -1.34,\ 1.49,\ 4.17).$$

Lengths 0.141 and 6.326: a 45-fold difference. **The weak direction is the coefficient change the predictions can't see**, which is why the data can't pin it down and why noise settles there.

Slide 23 Why The Small Eigenvalue Marks The Unstable Direction



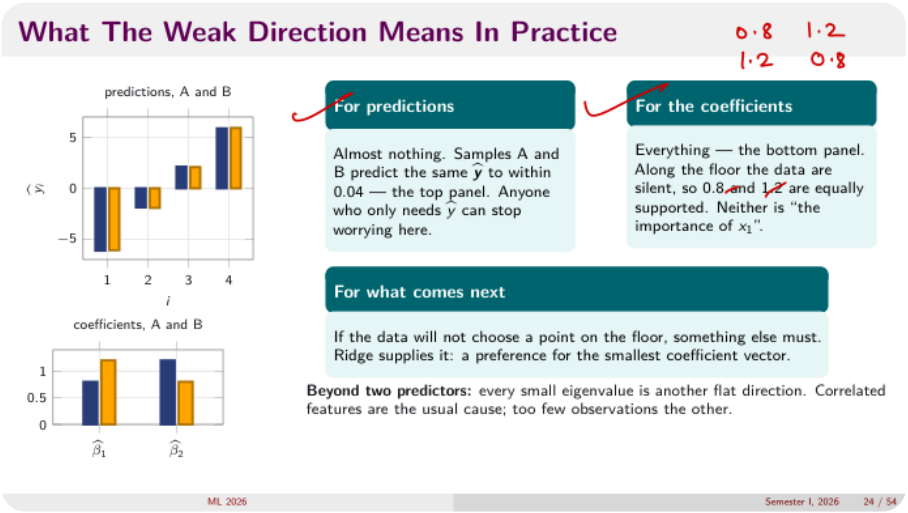
Slide 23 — Two parabolas from the minimum: ½(40.02)t² (the wall) and ½(0.02)t² (the floor).

$X^\top X q_j = \mu_j q_j \Rightarrow q_j^\top X^\top X q_j = \mu_j \Rightarrow \|Xq_j\|^2 = \mu_j$. Stepping t along q_j raises J by $\frac{1}{2}\mu_j t^2$. $\sqrt{0.01999} = 0.141$, $\sqrt{40.02} = 6.326$.

Curvature of the loss, size of the prediction change, and eigenvalue are the same number.

Slide 24

What The Weak Direction Means In Practice



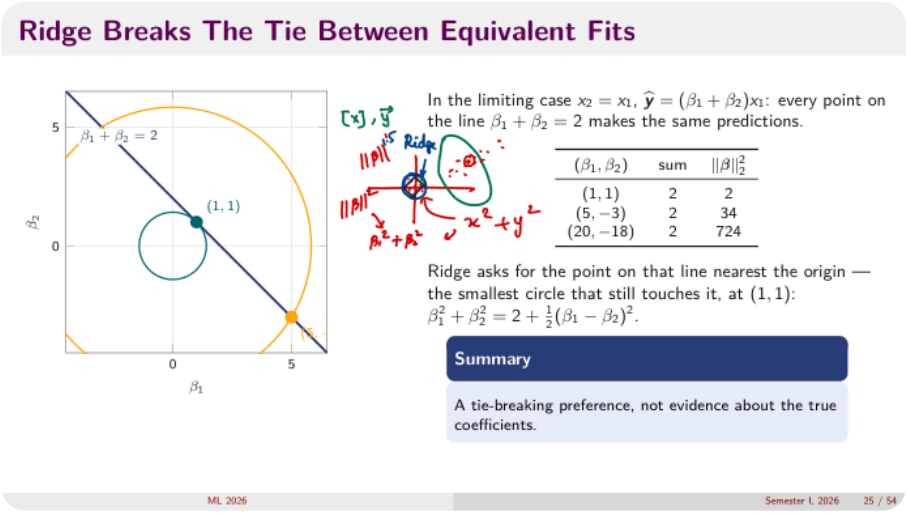
Slide 24 — Predictions from A and B agree to 0.04; coefficients swap 0.8 ↔ 1.2.

For predictions	For the coefficients
Almost nothing: A and B predict the same $\hat{y}$ to within 0.04. If you only need $\hat{y}$, stop worrying.	Everything: along the floor the data are silent, so 0.8 and 1.2 are equally supported. Neither is "the importance of x_1".

What comes next: if the data won't choose a point on the floor, something else must. Ridge supplies it: a preference for the smallest coefficient vector. Beyond 2 predictors: every small eigenvalue is another flat direction; correlated features are the usual cause, too few observations the other.

Slide 25

Ridge Breaks The Tie Between Equivalent Fits



Slide 25 — On the line ω₁ + ω₂ = 2, the smallest circle that touches it does so at (1, 1).

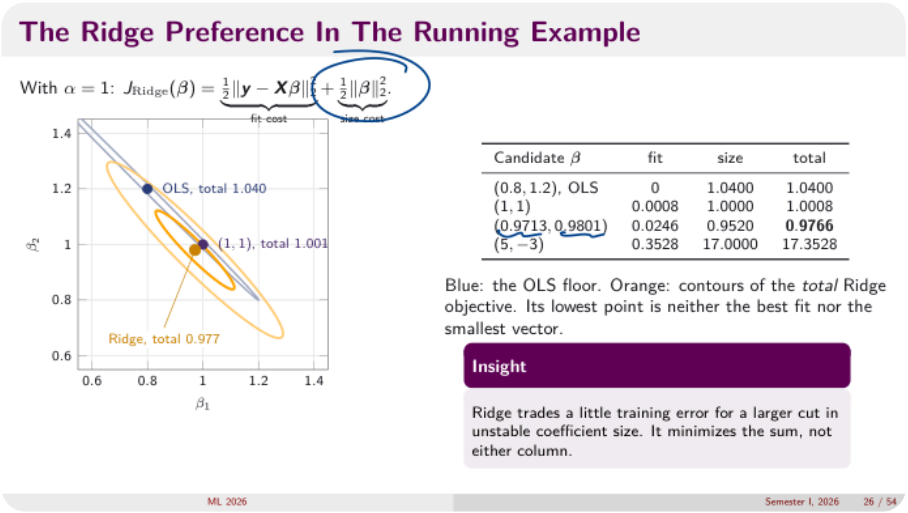
Limiting case $x_2 = x_1$: $\hat{y} = (\omega_1 + \omega_2)x_1$; every point on $\omega_1 + \omega_2 = 2$ makes the same predictions.

(ω_1, ω_2)	sum	$\|\omega\|^2$
$(1, 1)$	2	2
$(5, -3)$	2	34
$(20, -18)$	2	724

Identity: $\omega_1^2 + \omega_2^2 = \frac{1}{2}(\omega_1 + \omega_2)^2 + \frac{1}{2}(\omega_1 - \omega_2)^2 = 2 + \frac{1}{2}(\omega_1 - \omega_2)^2$ on the line. Minimised when $\omega_1 = \omega_2$, i.e. $(1, 1)$.

A tie-breaking preference, not evidence about the true coefficients.

Slide 26 The Ridge Preference In The Running Example



Slide 26 — Total ridge objective contours (orange) over the OLS floor (blue).

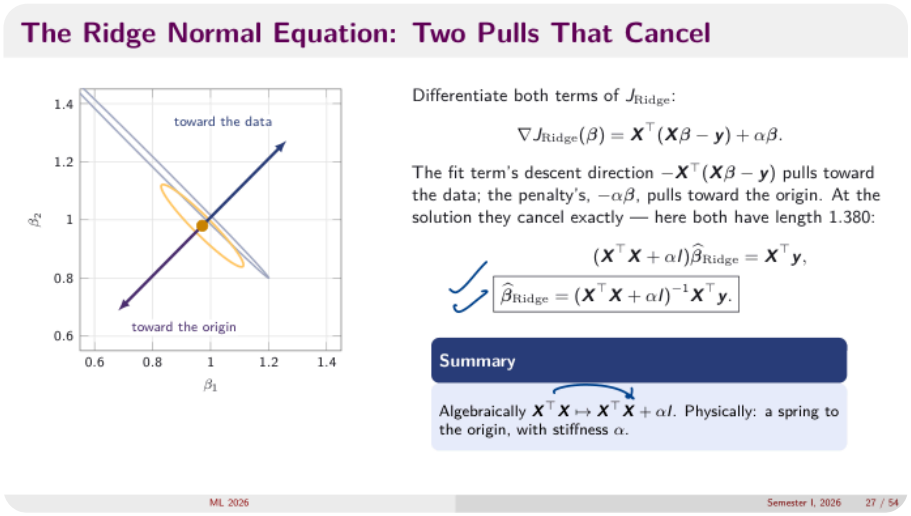
$$\lambda = 1\text{: } J_{\text{ridge}} = \underbrace{\tfrac{1}{2}\|y - X\omega\|^2}_{\text{fit}} + \underbrace{\tfrac{1}{2}\|\omega\|^2}_{\text{size}}.$$

candidate ω	fit	size	total
$(0.8, 1.2)$ OLS	0	$\frac{1}{2}(0.64 + 1.44) = 1.0400$	1.0400
$(1, 1)$	0.0008	1.0000	1.0008
$(0.9713, 0.9801)$ ridge	0.0246	0.9520	**0.9766**
$(5, -3)$	0.3528	17.0000	17.3528

The ridge minimum is neither the best fit nor the smallest vector. It minimises the **sum**: trades a little training error for a larger cut in unstable size.

Slide 27

The Ridge Normal Equation: Two Pulls That Cancel



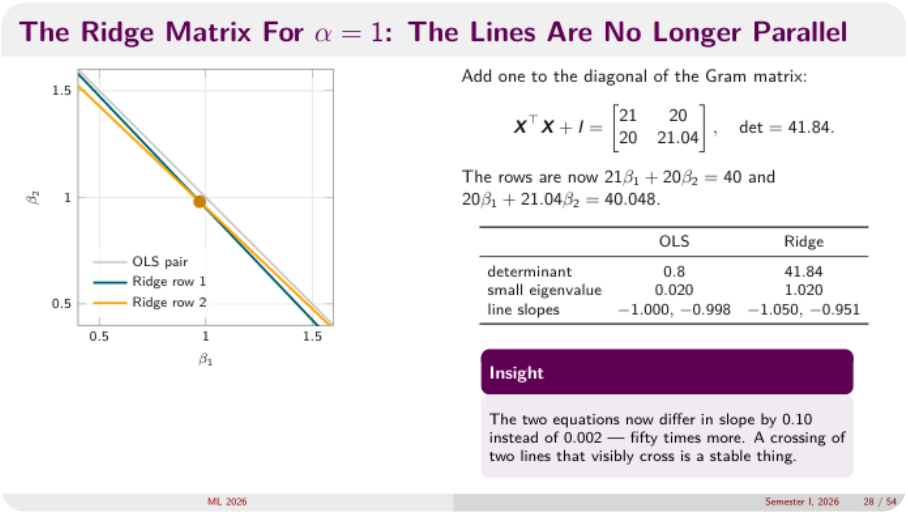
Slide 27 — At the ridge solution, the pull toward the data and the pull toward the origin have equal length (1.380) and cancel.

$$\nabla J_{\text{ridge}} = X^\top(X\omega - y) + \lambda\omega = 0 \;\Rightarrow\; (X^\top X + \lambda I)\hat{\omega}_{\text{ridge}} = X^\top y.$$

Descent directions: $-X^\top(X\omega - y)$ pulls toward the data; $-\lambda\omega$ pulls toward the origin. At the solution they're equal and opposite: $\|X^\top(y - X\hat{\omega})\| = \|\lambda\hat{\omega}\| = 1.380$ ✓.

Algebraically $X^\top X \to X^\top X + \lambda I$. Physically: a spring to the origin with stiffness λ.

Slide 28 The Ridge Matrix For $\lambda = 1$: The Lines Are No Longer Parallel



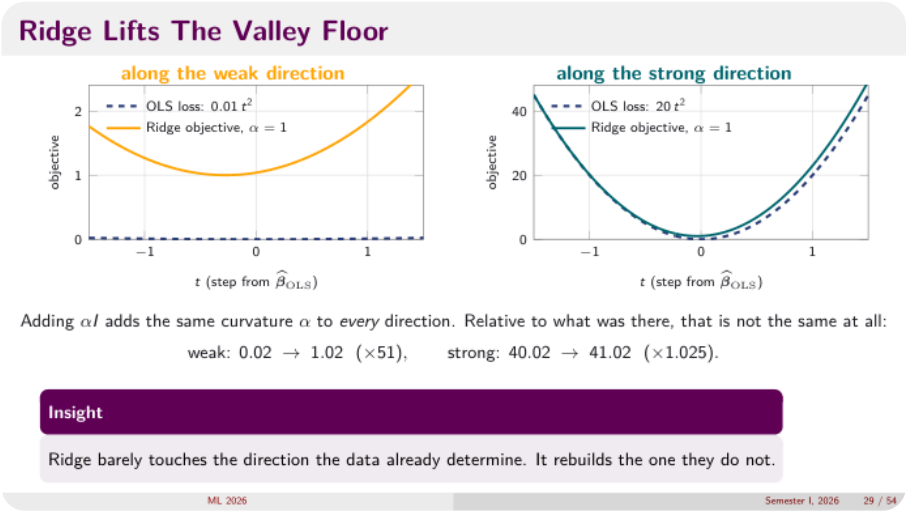
Slide 28 — Ridge rows $21\omega_1 + 20\omega_2 = 40$ and $20\omega_1 + 21.04\omega_2 = 40.048$ visibly cross.

$$X^\top X + I = \begin{bmatrix} 21 & 20 \\ 20 & 21.04 \end{bmatrix}, \det = 441.84 - 400 = 41.84.$$

	OLS	Ridge
determinant	0.8	41.84
small eigenvalue	0.020	1.020
line slopes	$-1.000, -0.998$	$-1.050, -0.951$

The slopes now differ by 0.10 instead of 0.002 (×50). **A crossing of two lines that visibly cross is a stable thing.**

Slide 29 Ridge Lifts The Valley Floor



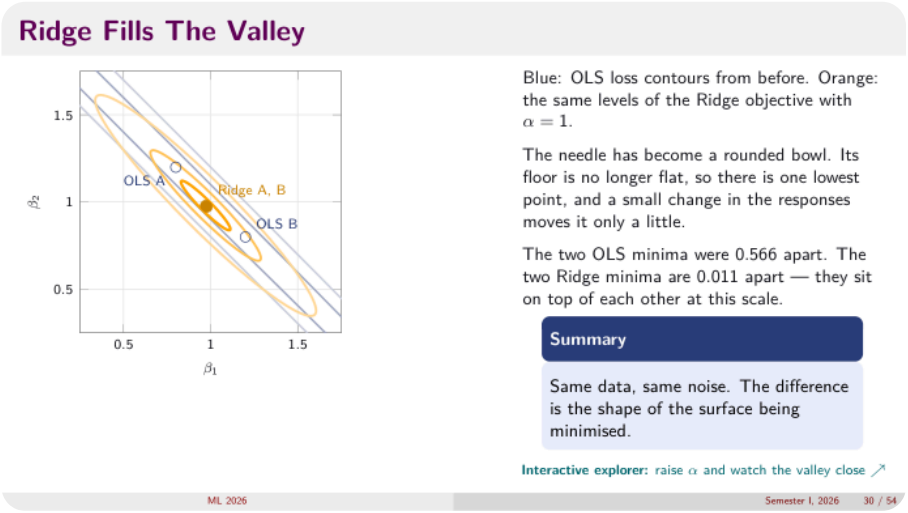
Slide 29 — Along the weak direction ridge turns 0.01t² into 0.51t²; along the strong direction 20t² barely changes.

λI adds the same curvature λ to every direction. Relative to what was there, that's very different:

$$\text{weak: } 0.02 \to 1.02 \ (\times 51), \qquad \text{strong: } 40.02 \to 41.02 \ (\times 1.025).$$

Ridge barely touches the direction the data already determine. It rebuilds the one they don't.

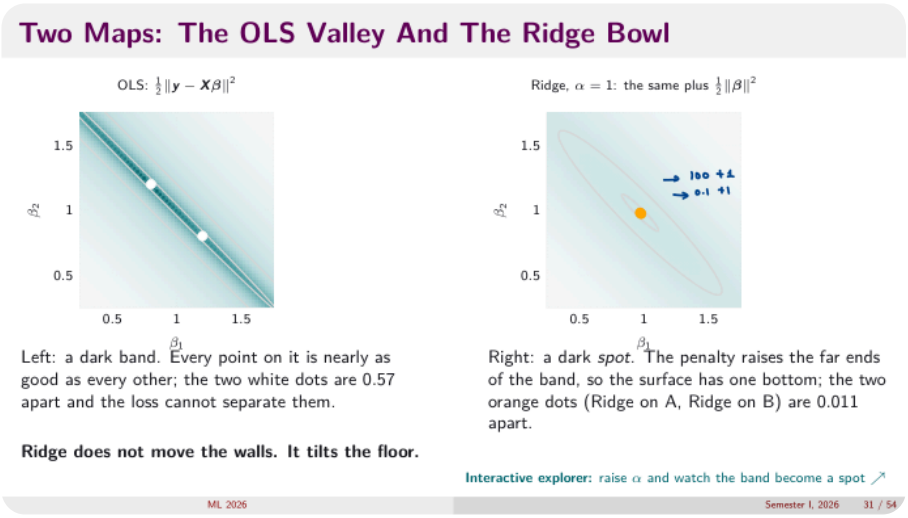
Slide 30 Ridge Fills The Valley



Slide 30 — Blue OLS contours vs orange ridge contours: the needle becomes a rounded bowl.

The ridge objective's level sets are a rounded bowl with a single lowest point. The two OLS minima were 0.566 apart; the two ridge minima are **0.011** apart.

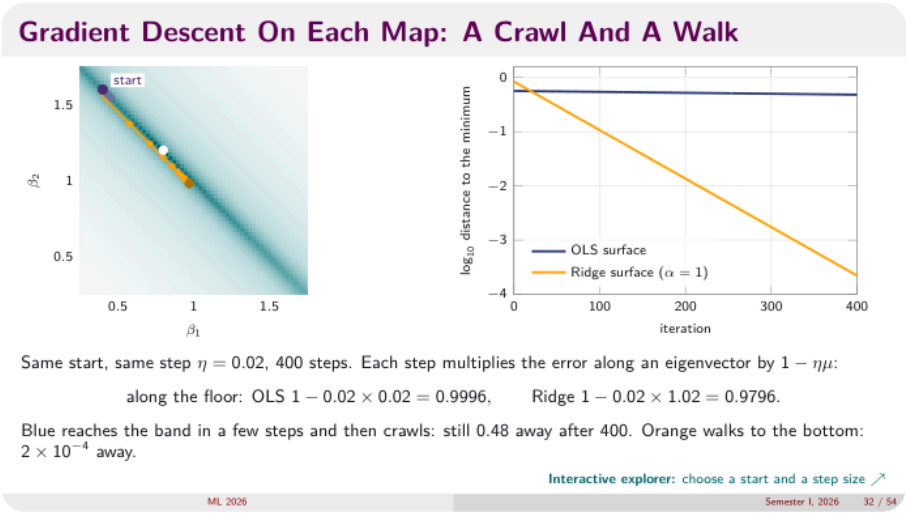
Slide 31 Two Maps: The OLS Valley And The Ridge Bowl



Slide 31 — Left: a dark band (OLS). Right: a dark spot (ridge).

Left: every point on the band is nearly as good as any other; the white dots are 0.57 apart and the loss can't separate them. Right: the penalty raises the far ends of the band so there's one bottom; orange dots 0.011 apart. **Ridge does not move the walls. It tilts the floor.**

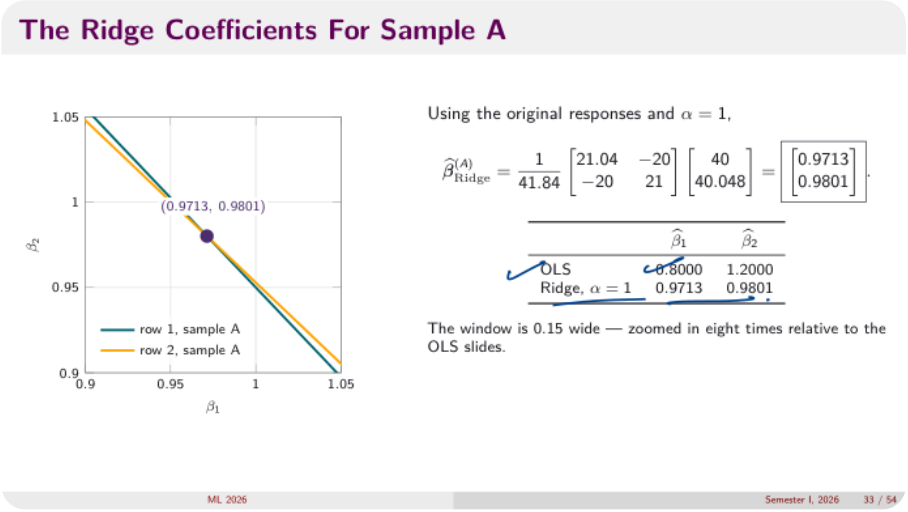
Slide 32 Gradient Descent On Each Map: A Crawl And A Walk



Slide 32 — Same start, η = 0.02, 400 steps: OLS still 0.48 away; ridge $2{\times}10^{-4}$ away.

Each step multiplies the error along an eigenvector by $1 - \eta\mu$ (A5): along the floor OLS $1 - 0.02 \times 0.02 = 0.9996$, ridge $1 - 0.02 \times 1.02 = 0.9796$. OLS reaches the band in a few steps, then crawls: still 0.48 away after 400. Ridge: 2×10^{-4}.

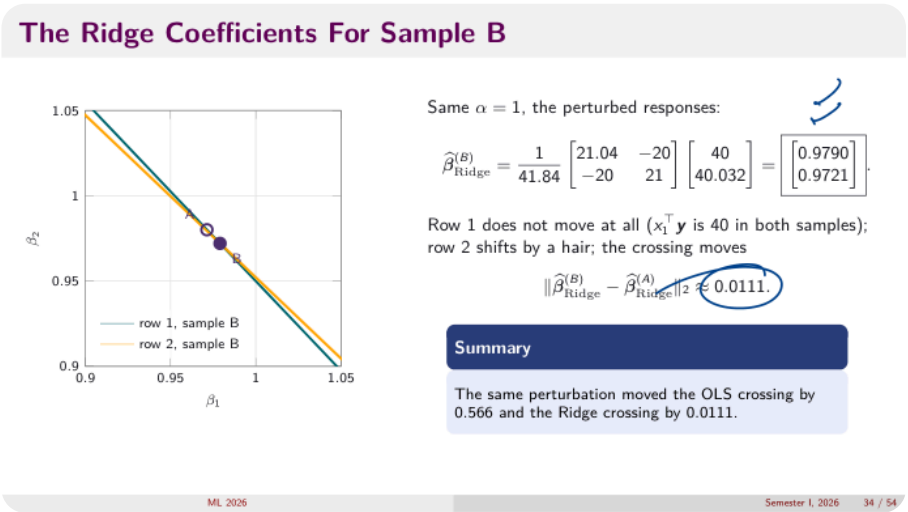
Slide 33 The Ridge Coefficients For Sample A



Slide 33 — Zoomed 8×: ridge rows for sample A cross at (0.9713, 0.9801).

$$\hat{\omega}^{(A)}_{\text{ridge}} = \frac{1}{41.84}\begin{bmatrix}21.04 & -20\\ -20 & 21\end{bmatrix}\begin{bmatrix}40\\ 40.048\end{bmatrix} = \frac{1}{41.84}\begin{bmatrix}841.6 - 800.96\\ -800 + 841.008\end{bmatrix} = \frac{1}{41.84}\begin{bmatrix}40.64\\ 41.008\end{bmatrix} = \begin{bmatrix}0.9713\\ 0.9801\end{bmatrix}.$$

Slide 34 The Ridge Coefficients For Sample B

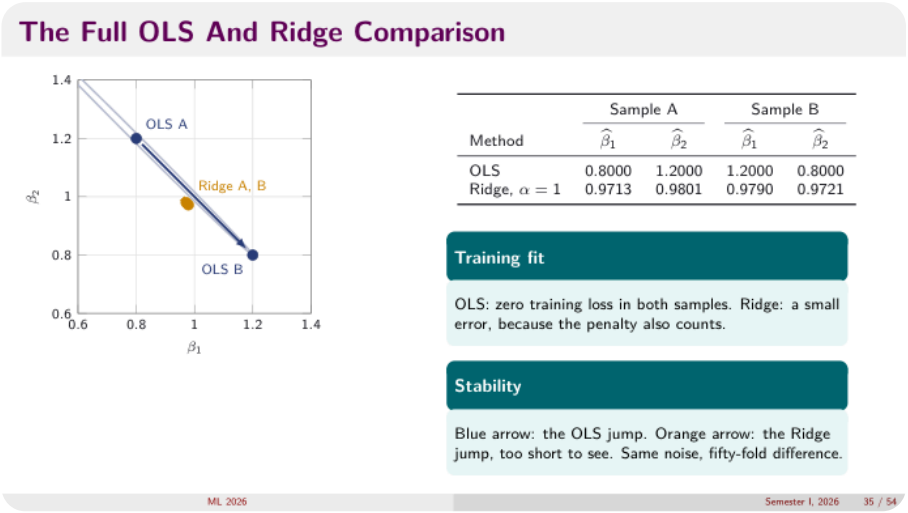


Slide 34 — Ridge rows for sample B: row 1 doesn't move; row 2 shifts by a hair.

$$\hat{\omega}_{\text{ridge}}^{(B)} = \frac{1}{41.84}\begin{bmatrix}21.04(40) - 20(40.032)\\ -20(40) + 21(40.032)\end{bmatrix} = \frac{1}{41.84}\begin{bmatrix}40.96\\ 40.672\end{bmatrix} = \begin{bmatrix}0.9790\\ 0.9721\end{bmatrix}.$$

Row 1 doesn't move ($x_1^\top y = 40$ in both). Jump $= \sqrt{0.0077^2 + 0.0080^2} = 0.0111$. **The same perturbation moved OLS by 0.566 and ridge by 0.0111.**

Slide 35 The Full OLS And Ridge Comparison



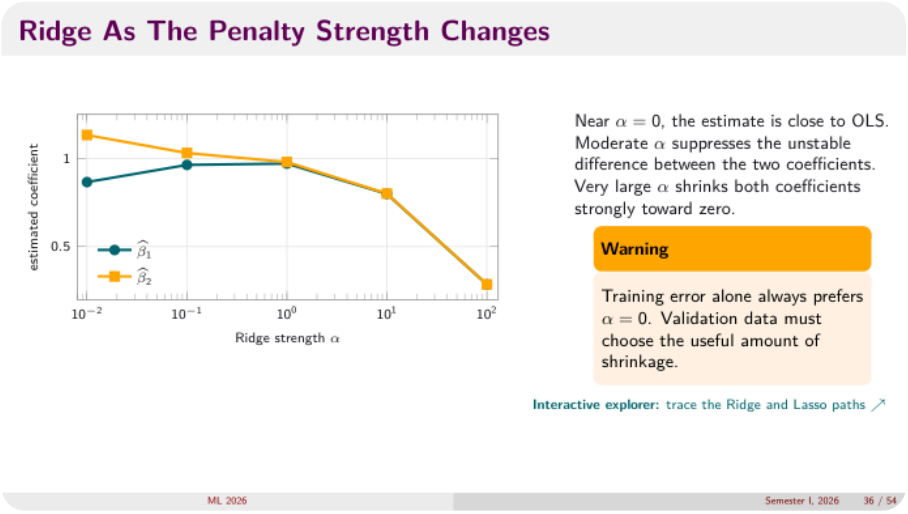
Slide 35 — Blue arrow: the OLS jump. Orange arrow: the ridge jump, too short to see.

method	A: $\hat{\omega}_1$	A: $\hat{\omega}_2$	B: $\hat{\omega}_1$	B: $\hat{\omega}_2$
OLS	0.8000	1.2000	1.2000	0.8000
Ridge, $\lambda = 1$	0.9713	0.9801	0.9790	0.9721

Training fit: OLS zero loss in both; ridge a small error (the penalty also counts). **Stability:** fifty-fold difference.

Slide 36

Ridge As The Penalty Strength Changes

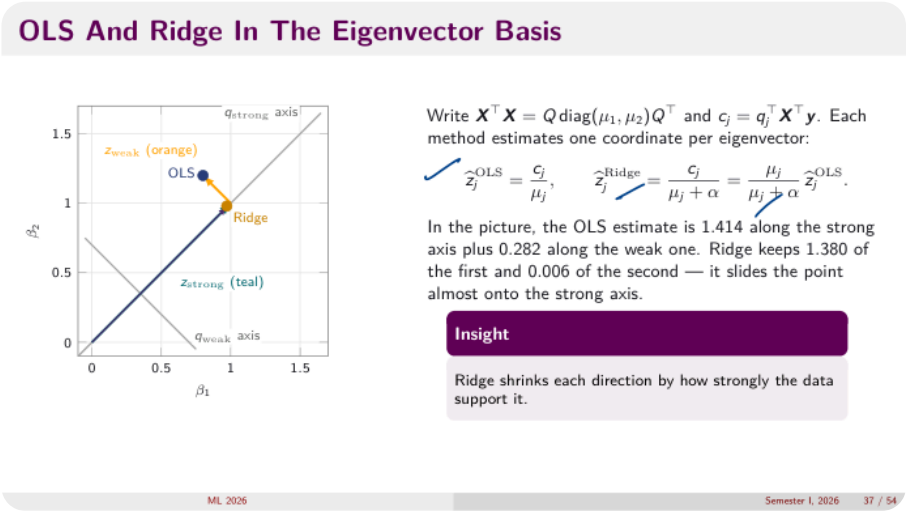


Slide 36 — Ridge path vs λ: near 0 it is OLS; moderate λ merges the two coefficients; large λ shrinks both.

λ	$\hat{\omega}^{(A)}$
0	$(0.800, 1.200)$
0.06	$(0.948, 1.049)$
0.1	$(0.964, 1.031)$
1	$(0.971, 0.980)$
10	$(0.799, 0.801)$

Careful. Training error alone always prefers $\lambda = 0$. Validation must choose the useful amount (slide 52).

Slide 37 OLS And Ridge In The Eigenvector Basis

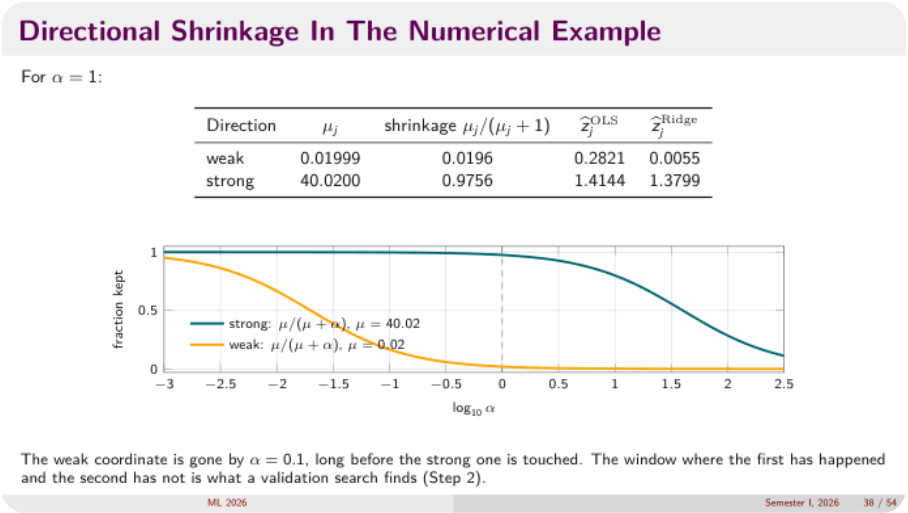


Slide 37 — OLS = 1.414 along strong + 0.282 along weak; ridge keeps 1.380 and 0.006.

$c_j = q_j^\top X^\top y$; $\hat{z}_j^{\mathrm{OLS}} = c_j/\mu_j$, $\hat{z}_j^{\mathrm{ridge}} = c_j/(\mu_j + \lambda) = \frac{\mu_j}{\mu_j+\lambda}\hat{z}_j^{\mathrm{OLS}}$.

$c_{\text{strong}} = 56.60$, $c_{\text{weak}} = 0.0056$. OLS: $56.60/40.02 = 1.414$ and $0.0056/0.01999 = 0.282$. Ridge: $56.60/41.02 = 1.380$ and $0.0056/1.01999 = 0.0055$. **It slides the point almost onto the strong axis**, i.e. onto $\omega_1 \approx \omega_2$.

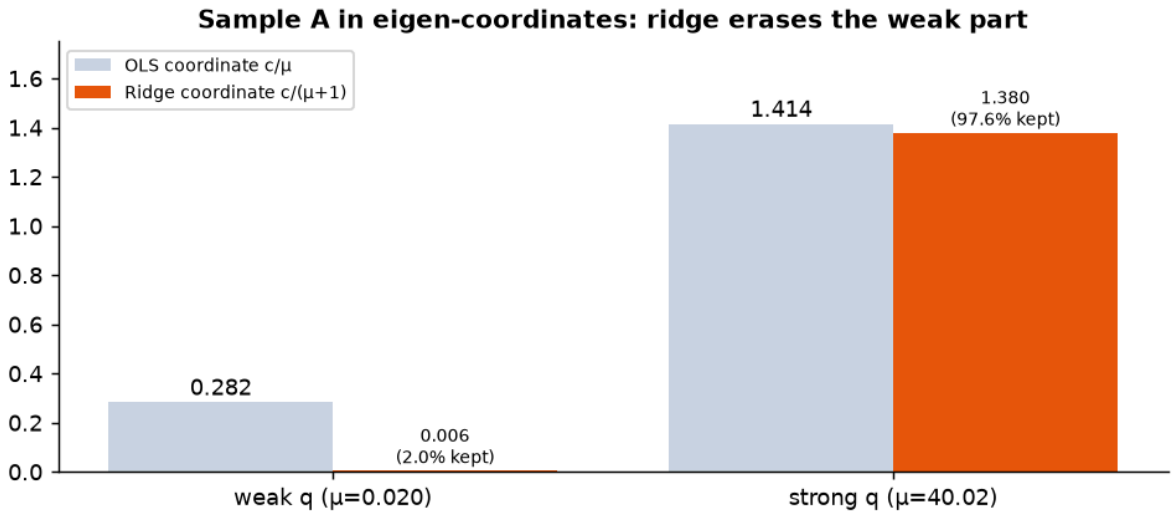
Slide 38 Directional Shrinkage In The Numerical Example



Slide 38 — Fraction kept vs log λ: the weak coordinate is gone by λ = 0.1, long before the strong one is touched.

$\lambda = 1$:

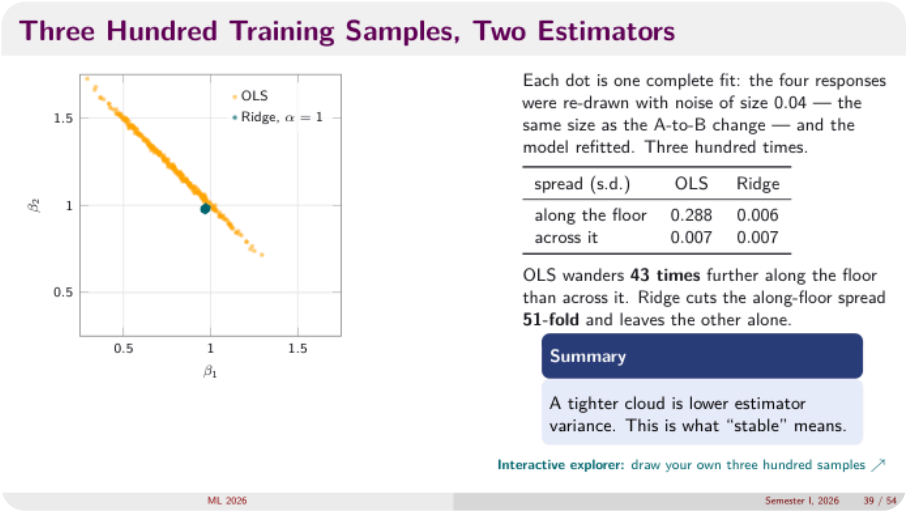
direction	μ_j	shrinkage $\mu_j/(\mu_j+1)$	$\hat{z}^{\mathrm{OLS}}$	$\hat{z}^{\mathrm{ridge}}$
weak	0.01999	0.0196	0.2821	0.0055
strong	40.02	0.9756	1.4144	1.3799



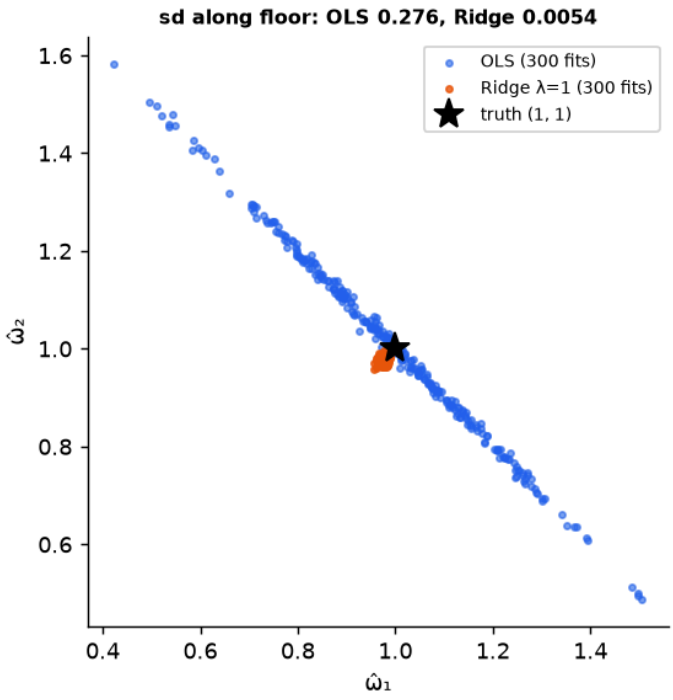
Sample A in eigen-coordinates: ridge keeps 97.6% of the strong part and 2% of the weak part.

The window where the weak coordinate is gone and the strong one untouched (roughly $0.1 \lesssim \lambda \lesssim 1$ here) is what validation finds.

Slide 39 Three Hundred Training Samples, Two Estimators



Slide 39 — 300 refits with noise 0.04: the OLS cloud is a long streak along the floor; the ridge cloud is a dot.



My own 300-sample run (truth (1, 1), noise sd 0.04): same picture, sd along the floor 0.28 for OLS vs 0.005 for ridge.

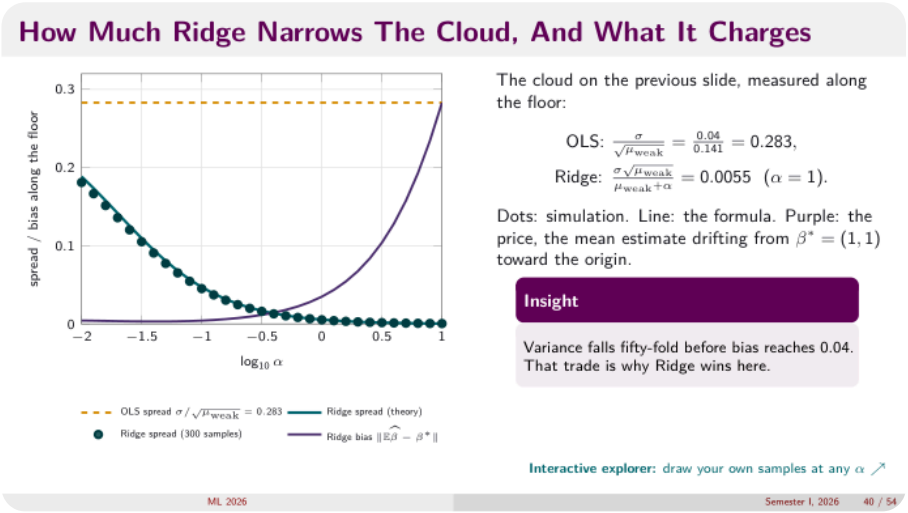
spread (sd)	OLS	Ridge
along the floor	$\sigma/\sqrt{\mu_w} = 0.04/0.141 = 0.283$	$\sigma\sqrt{\mu_w}/(\mu_w+1) = 0.0055$
across it	$0.04/6.326 = 0.0063$	$0.04(6.326)/41.02 = 0.0062$

OLS wanders ~45× further along the floor than across. Ridge cuts the along-floor spread **51-fold** and leaves the other alone. **A tighter cloud is lower estimator variance. That's what "stable" means.**

(Derivation: $\hat{z}_j = c_j/\mu_j$, and $c_j = q_j^\top X^\top y$ has noise sd $\sigma\|Xq_j\| = \sigma\sqrt{\mu_j}$; divide by μ_j or by $\mu_j + \lambda$.)

Slide 40

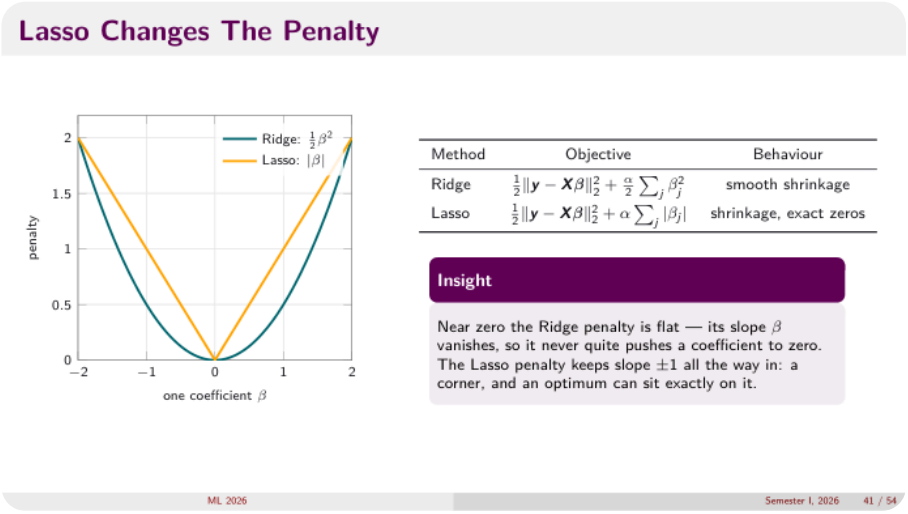
How Much Ridge Narrows The Cloud, And What It Charges



Slide 40 — Spread falls fifty-fold before the bias (purple) reaches 0.04.

OLS along-floor spread 0.283; ridge 0.0055 at $\lambda = 1$. Purple curve = the price: the mean estimate drifting from $\omega^* = (1, 1)$ toward 0, i.e. bias $= \lambda/(\mu_s + \lambda)\times$ the strong coordinate $\approx \frac{1}{41} \cdot 1.414 = 0.034$. **Variance falls fifty-fold before bias reaches 0.04. That trade is why ridge wins here.**

Slide 41 Lasso Changes The Penalty

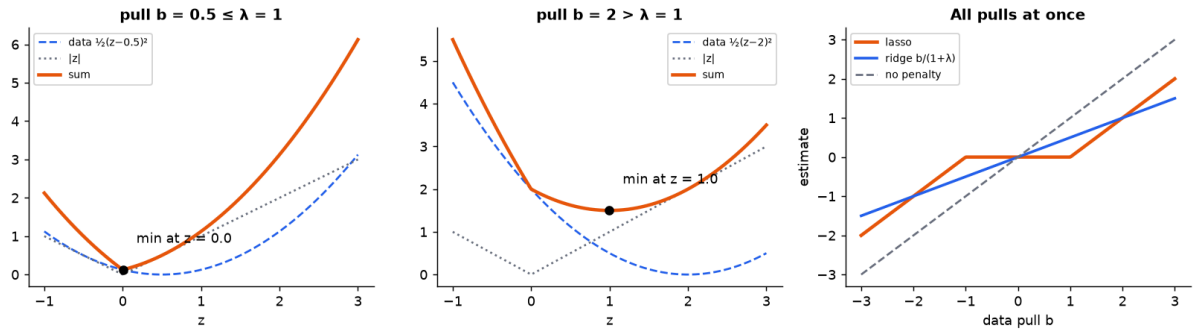


Slide 41 — Ridge ½ω² is flat at 0; lasso |ω| has a corner.

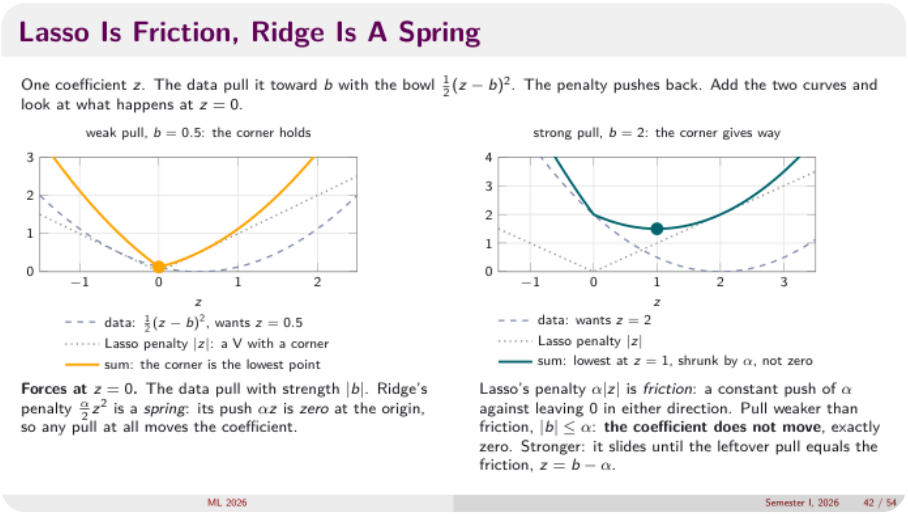
method	objective	behaviour
Ridge	$\frac{1}{2}\|y - X\omega\|^2 + \frac{\lambda}{2}\sum \omega_j^2$	smooth shrinkage
Lasso	$\frac{1}{2}\|y - X\omega\|^2 + \lambda\sum \|\omega_j\|$	shrinkage, exact zeros

Near 0 the ridge penalty's slope $\lambda\omega$ vanishes, so it never quite pushes to 0. Lasso keeps slope $\pm\lambda$ all the way in: a **corner**, and an optimum can sit exactly on it.

Slide 42 Lasso Is Friction, Ridge Is A Spring



Three steps on one coefficient (λ = 1): a weak pull b = 0.5 leaves the minimum at the corner z = 0; a strong pull b = 2 moves it to z = 1; the full input–output rule.



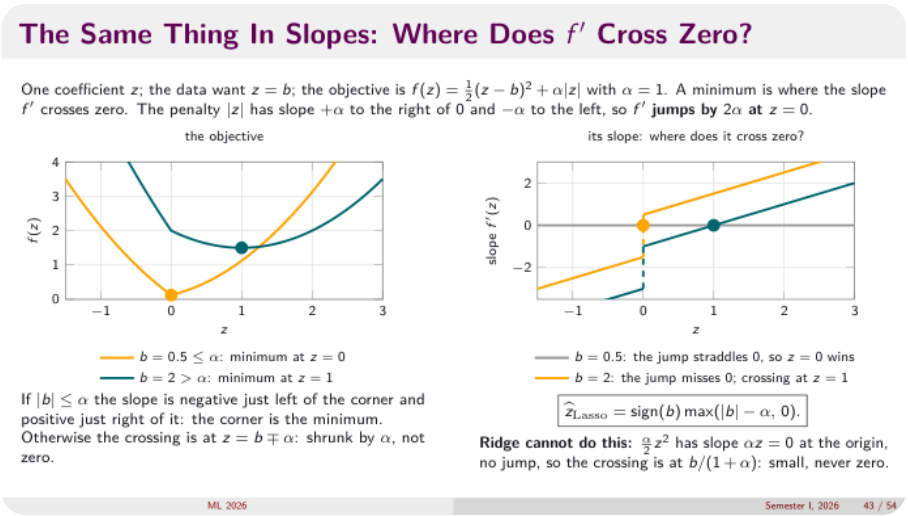
Slide 42 — Data bowl ½(z − b)² + lasso V: sum's lowest point.

One coefficient z; data pull it toward b via $\frac{1}{2}(z-b)^2$. **Forces at $z=0$:** data pull with strength $|b|$.

- Ridge $\frac{\lambda}{2}z^2$ is a **spring**: push λz is zero at the origin, so any pull moves the coefficient.
- Lasso $\lambda|z|$ is **friction**: a constant push λ against leaving 0. If $|b| \leq \lambda$: doesn't move, exactly 0. Stronger: slides until leftover pull equals friction, $z = b - \lambda$.

$b = 0.5$, $\lambda = 1$: corner holds, $z = 0$. $b = 2$: $z = 1$.

Slide 43 The Same Thing In Slopes: Where Does f' Cross Zero?



Slide 43 — The slope f′ jumps by 2λ at z = 0; if the jump straddles 0, the corner is the minimum.

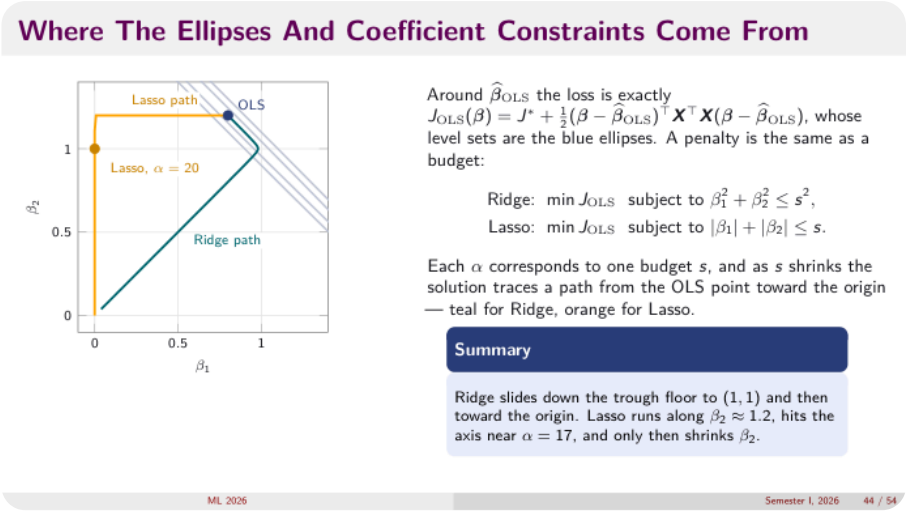
$f(z) = \frac{1}{2}(z-b)^2 + \lambda|z|$. Right of 0: $f'(z) = z - b + \lambda$; left of 0: $f'(z) = z - b - \lambda$. At $z = 0$ the slope jumps from $-b - \lambda$ to $-b + \lambda$ (a jump of 2λ).

- If $|b| \le \lambda$: left slope ≤ 0, right slope ≥ 0 → the corner is the minimum.
- Otherwise the crossing is at $z = b \mp \lambda$.

$$\hat{z}_{\text{lasso}} = \text{sign}(b)\max(|b| - \lambda, 0).$$

Ridge can't do this: $\frac{\lambda}{2}z^2$ has slope 0 at the origin, no jump, so the crossing is $b/(1+\lambda)$: small, never 0.

Slide 44 Where The Ellipses And Coefficient Constraints Come From



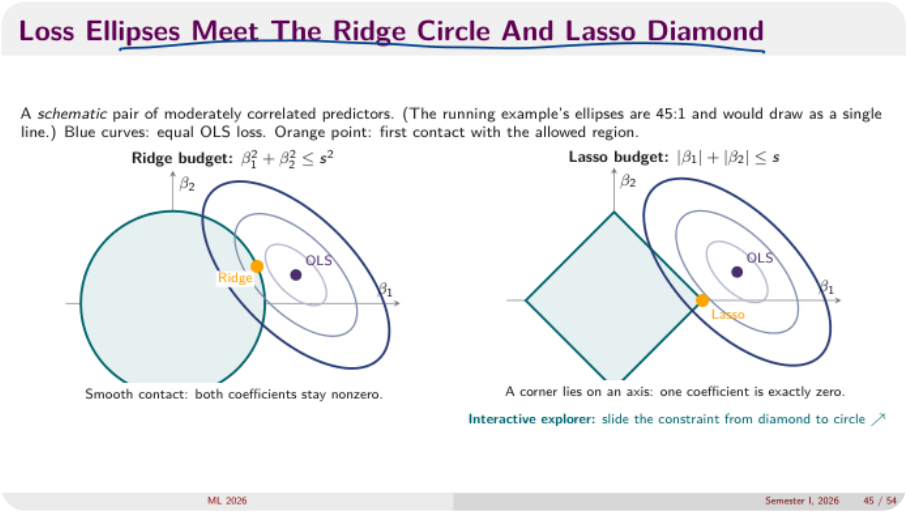
Slide 44 — Ridge path slides down the floor to (1, 1) then toward 0; lasso path runs along $\omega_2 \approx 1.2$, hits the axis, then shrinks ω_2.

Around $\hat{\omega}_{\text{OLS}}$ the loss is exactly $J^* + \frac{1}{2}(\omega - \hat{\omega})^\top X^\top X(\omega - \hat{\omega})$: the blue ellipses. Penalty ⇔ budget:

$$\text{Ridge: } \min J \text{ s.t. } \omega_1^2 + \omega_2^2 \le s^2, \qquad \text{Lasso: } \min J \text{ s.t. } |\omega_1| + |\omega_2| \le s.$$

As s shrinks, the solution traces a path to the origin. Lasso runs along $\omega_2 = 1.2$ (exactly, A6 Phase 1), hits the axis at $\lambda = 16$, then shrinks ω_2.

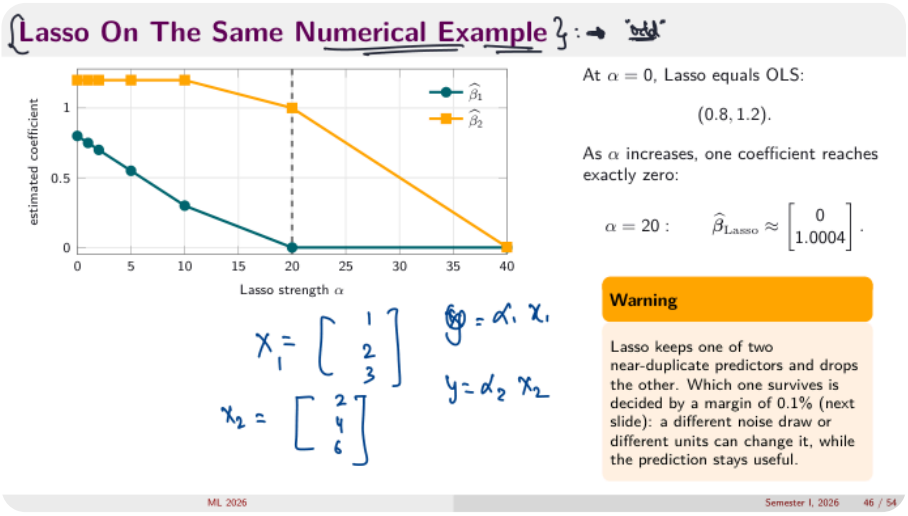
Slide 45 Loss Ellipses Meet The Ridge Circle And Lasso Diamond



Slide 45 — Schematic, moderately correlated: smooth contact with the circle, corner contact with the diamond.

(Schematic, since the running example's 45:1 ellipses would draw as a line.) First contact of the loss ellipses with the circle is smooth → both nonzero. With the diamond it's at a corner on an axis → one coefficient exactly 0 (L7 A8).

Slide 46 Lasso On The Same Numerical Example



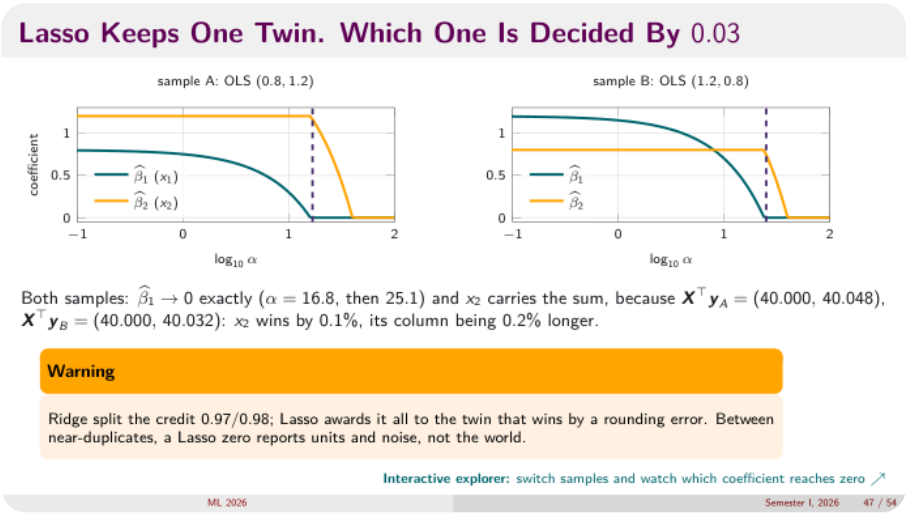
Slide 46 — Lasso path vs λ: at λ = 20, ω̂ ≈ (0, 1.0004).

$\lambda = 0$: OLS $(0.8, 1.2)$. As λ grows, ω_1 reaches exactly 0. At $\lambda = 20$: $\hat{\omega}_{\text{lasso}} = (0,\ 1.0004)$ (A6 Phase 2: $(40.048 - 20)/20.04$).

Careful. Lasso keeps one twin and drops the other; which one survives is decided by a margin of 0.1%. A different noise draw or different units can change it, while the prediction stays useful. (Annotation: $\hat{y} = \omega_2 x_2$ with $\omega_2 \approx 1$; that's half the true combined effect of 2, because the penalty also shrinks it.)

Slide 47

Lasso Keeps One Twin. Which One Is Decided By 0.03

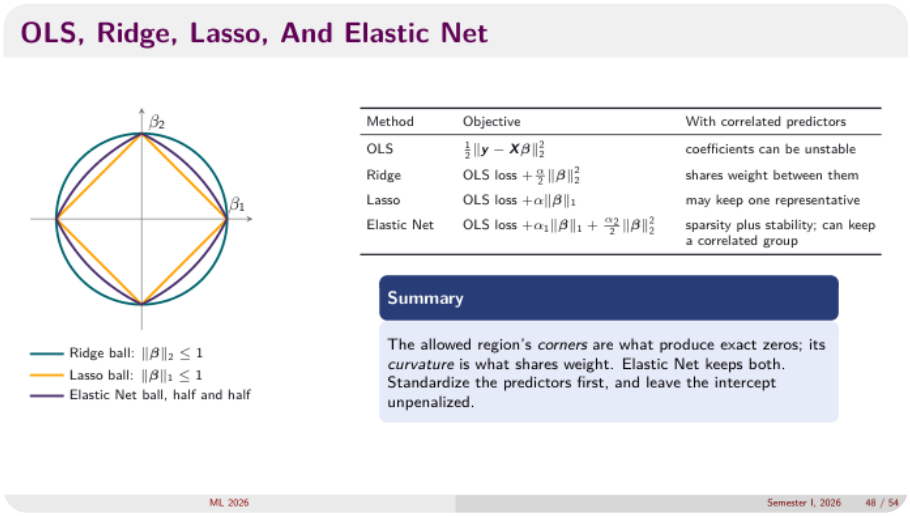


Slide 47 — In both samples ω_1 hits zero first and x_2 carries the sum.

Both samples: $\hat{\omega}_1 = 0$ first (exact kinks $\lambda = 16$ for A, 24 for B; the slide reads 16.8, 25.1 off its grid), and x_2 carries the sum. Because $X^\top y_A = (40.000, 40.048)$ and $X^\top y_B = (40.000, 40.032)$: x_2 wins by 0.1%, its column being 0.2% longer.

Careful. Ridge split the credit 0.97/0.98; lasso awards it all to the twin that wins by a rounding error. **Between near-duplicates, a lasso zero reports units and noise, not the world.**

Slide 48 OLS, Ridge, Lasso, And Elastic Net



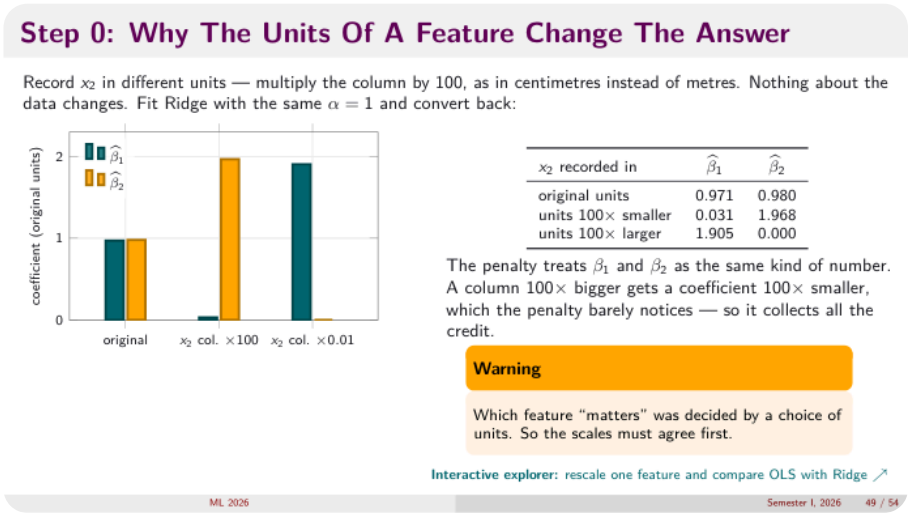
Slide 48 — Unit balls: ℓ2 circle, ℓ1 diamond, elastic net (half-and-half) rounded diamond.

method	objective	with correlated predictors
OLS	$\frac{1}{2}\|y - X\omega\|^2$	coefficients can be unstable
Ridge	OLS loss + $\frac{\lambda}{2}\|\omega\|^2$	shares weight between them
Lasso	OLS loss + $\lambda\|\omega\|_1$	may keep one representative
Elastic net	OLS loss + $\lambda_1\|\omega\|_1 + \frac{\lambda_2}{2}\|\omega\|^2$	sparsity plus stability; can keep a correlated group

The allowed region's corners produce exact zeros; its curvature shares weight. Elastic net keeps both. Standardise first; leave the intercept unpenalised.

Slide 49

Step 0: Why The Units Of A Feature Change The Answer

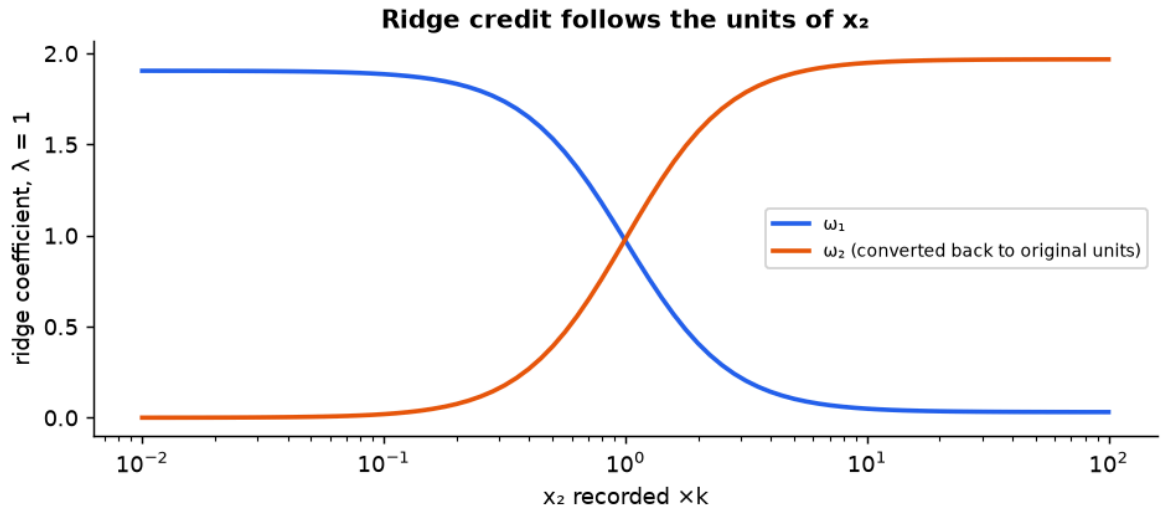


Slide 49 — Recording x_2 ×100 or ×0.01 hands all the ridge credit to one feature.

Ridge $\lambda = 1$, converted back to original units:

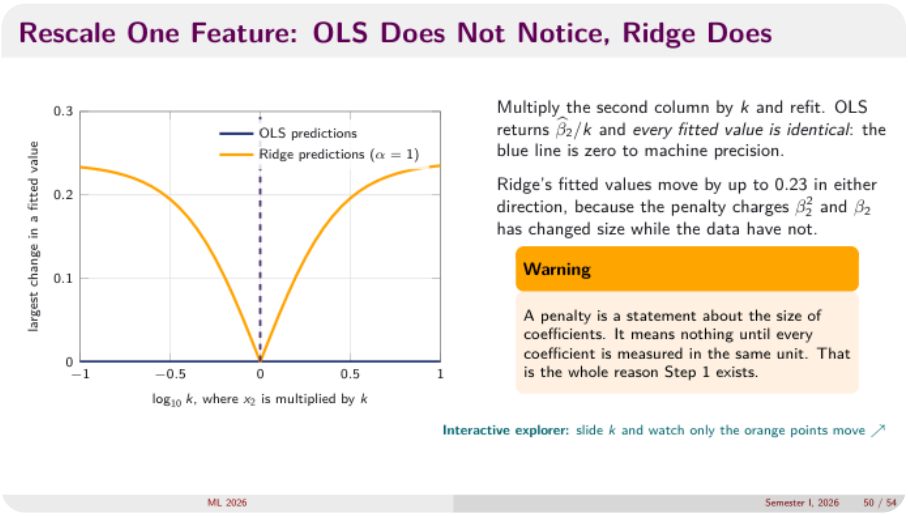
x_2 recorded in	$\hat{\omega}_1$	$\hat{\omega}_2$
original units	0.971	0.980
×100 (units 100× smaller)	0.031	1.968
×0.01 (units 100× larger)	1.905	0.0002

The penalty treats ω_1, ω_2 as the same kind of number. A column 100× bigger needs a coefficient 100× smaller, which the penalty barely notices, so it **collects all the credit**.



Sweeping the scale k of x_2: ridge credit flips from x_1 to x_2 purely because of units.

Slide 50 Rescale One Feature: OLS Does Not Notice, Ridge Does



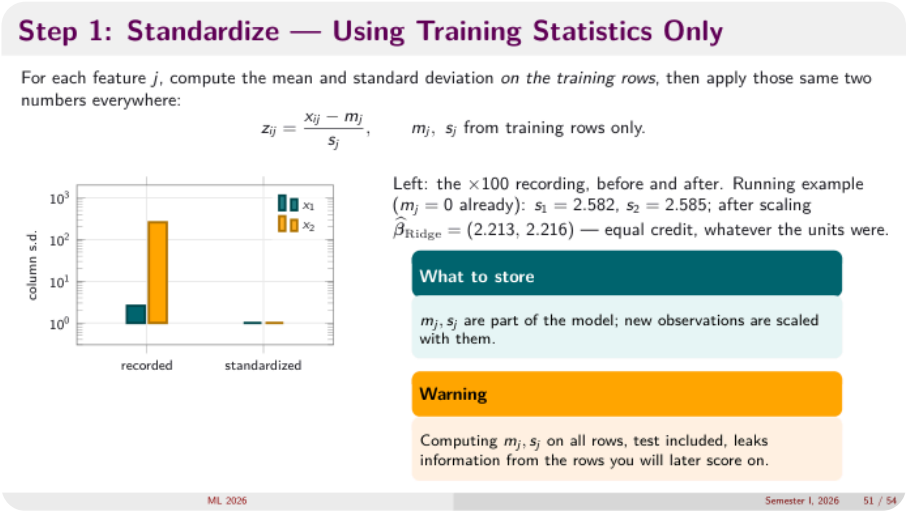
Slide 50 — Largest change in a fitted value vs k: zero for OLS, up to 0.24 for ridge.

Multiply column 2 by k and refit. OLS returns $\hat{\omega}_2/k$; every fitted value identical (machine precision). Ridge's fitted values move by up to 0.24, because the penalty charges ω_2^2 and ω_2 has changed size while the data have not.

Careful. A penalty means nothing until every coefficient is measured in the same unit.

Slide 51

Step 1: Standardize — Using Training Statistics Only



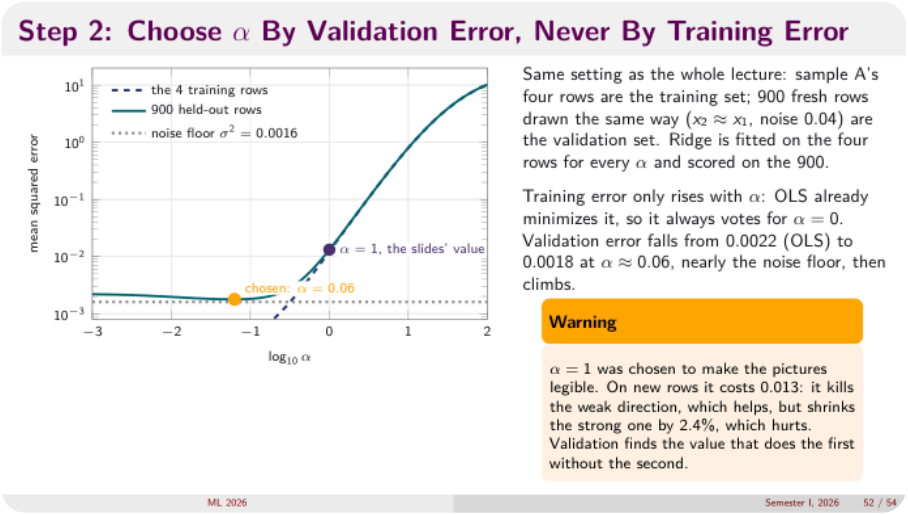
Slide 51 — After scaling, column sds agree and ridge gives equal credit.

$z_{ij} = (x_{ij} - m_j)/s_j$ with m_j, s_j from **training rows only**. Running example ($m_j = 0$): $s_1 = \sqrt{20/3} = 2.582$, $s_2 = \sqrt{20.04/3} = 2.585$. After scaling, $\hat{\omega}_{\text{ridge}} = (2.213, 2.216)$: equal credit, whatever the units were. (In standardised units, so ≈ 2.58× the raw coefficients.)

Store m_j, s_j as part of the model; scale new observations with them.

Careful. Computing m_j, s_j on all rows (test included) leaks information.

Slide 52 Step 2: Choose λ By Validation Error, Never By Training Error



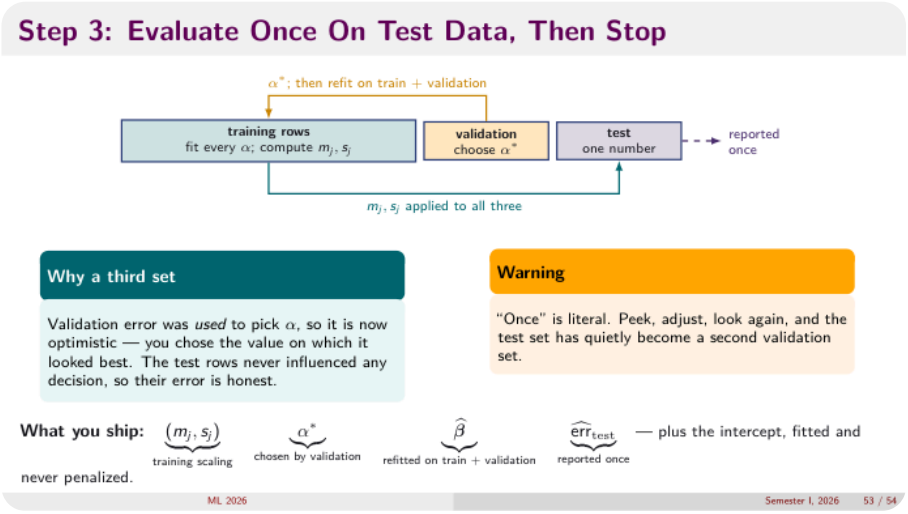
Slide 52 — Training MSE rises with λ; validation MSE on 900 fresh rows dips to ≈0.0018 at λ ≈ 0.06.

Sample A's four rows = training; 900 fresh rows drawn the same way ($x_2 \approx x_1$, noise 0.04) = validation. Ridge fitted on the 4 rows for each λ, scored on the 900.

- Training error only rises with λ → always votes for 0.
- Validation error: 0.0022 (OLS) → **0.0018 at** $\lambda \approx 0.06$ (near the noise floor $\sigma^2 = 0.0016$) → climbs.

Careful. $\lambda = 1$ was chosen to make pictures legible. On new rows it costs 0.013: it kills the weak direction (helps) but shrinks the strong one by 2.4% (hurts). Validation finds the value that does the first without the second.

Slide 53 Step 3: Evaluate Once On Test Data, Then Stop

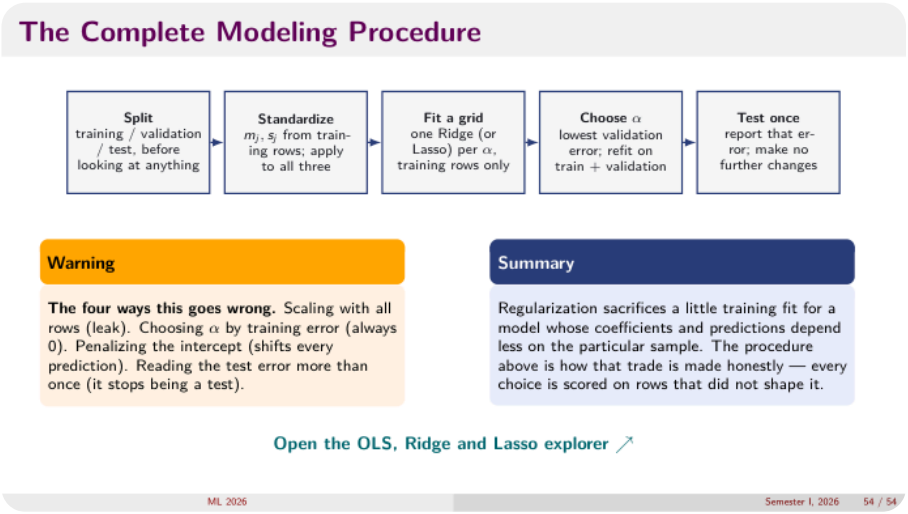


Slide 53 — training rows (fit every λ; compute m, s) → validation (choose λ) → test (one number).*

Validation error is now optimistic (you picked the value it liked). The test rows never influenced a decision, so their error is honest. **"Once" is literal**: peek, adjust, look again, and the test set has become a second validation set.

What you ship: (m_j, s_j) from training, λ^* from validation, $\hat{\omega}$ refitted on train + validation, $\mathrm{err}_{\mathrm{test}}$ reported once, plus the intercept (fitted, never penalised).

Slide 54 The Complete Modeling Procedure



Slide 54 — Split → standardise → fit a grid → choose λ → test once.

1. **Split** train / validation / test before looking at anything.
2. **Standardise** with training m_j, s_j; apply to all three.
3. **Fit a grid**: one ridge (or lasso) per λ, training rows only.
4. **Choose** λ: lowest validation error; refit on train + validation.
5. **Test once**: report that error; no further changes.

Careful. **Four ways it goes wrong:** scaling with all rows (leak); choosing λ by training error (always 0); penalising the intercept (shifts every prediction); reading the test error more than once.

Summary: regularisation sacrifices a little training fit for a model whose coefficients and predictions depend less on the particular sample; the procedure above makes that trade honestly.

Also worth knowing

Why each direction converges at its own rate. In eigen-coordinates, one step of gradient descent multiplies coordinate j by the **contraction factor** $1 - \eta\mu_j$. The step is stable when every factor is smaller than one in size, so $\eta < 2/\mu_{\max}$. A negative factor means zig-zag, and the flattest direction sets the overall speed.

Part C Practice

1. Verify $X^\top X$ and $X^\top y^{(A)}$ for the running dataset entry by entry.
2. Solve the OLS normal equation for sample B by subtracting the two equations (A2 method).
3. Compute the eigenvalues of $\begin{bmatrix} 20 & 20 \\ 20 & 20.04 \end{bmatrix}$ with the trace/determinant shortcut. What is the condition number?
4. Show $J_A(1.2, 0.8) = 0.0032$. Why is this exactly $\frac{1}{2}\|y^{(A)} - y^{(B)}\|^2$?
5. Compute ridge for sample A with $\lambda = 10$. Compare with the slide-36 table.
6. Show that in the eigenbasis, ridge's coefficient-jump between samples along the weak direction is $\frac{\mu_w}{\mu_w + \lambda}$ times OLS's. Evaluate for $\lambda = 1$.
7. Gradient descent with $\eta = 0.04$ on the OLS surface: what are the per-step factors along both eigenvectors? Is it stable? What about $\eta = 0.06$?
8. Lasso, sample A: give $\hat{\omega}$ at $\lambda = 10$ and at $\lambda = 30$ using A6.
9. For $\lambda = 1$ and one coefficient, give ridge and lasso estimates for $b = 0.7$ and $b = 3$.
0. If x_1 were recorded in units 10× larger (column ×0.1), what would OLS return? What would ridge tend to do?
1. Why must m_j, s_j be stored with the model?
2. Name the four procedural mistakes from slide 54 and the symptom each causes.

1. See A1 (20, 20, 20.04; 40, 40.048).
2. $20\omega_1 + 20\omega_2 = 40$, $20\omega_1 + 20.04\omega_2 = 40.032 \Rightarrow 0.04\omega_2 = 0.032 \Rightarrow \omega_2 = 0.8$, $\omega_1 = 1.2$.
3. $T = 40.04$, $D = 0.8$: $\mu = 20.02 \pm \sqrt{400.8004 - 0.8} = 20.02 \pm 20.0000 \Rightarrow 40.02, 0.02$ (more precisely 0.01999). $\kappa \approx 2002$.
4. $X(1.2, 0.8) = y^{(B)}$ (perfect fit for B), so residual on A is $y^{(A)} - y^{(B)} = (-0.04, 0.04, 0.04, -0.04)$; $\frac{1}{2}(4 \times 0.0016) = 0.0032$.
5. $\begin{bmatrix} 30 & 20 \\ 20 & 30.04 \end{bmatrix}^{-1}(40, 40.048)$: det $= 901.2 - 400 = 501.2$; $\frac{1}{501.2}(30.04 \cdot 40 - 20 \cdot 40.048,\ -800 + 30 \cdot 40.048) = \frac{1}{501.2}(400.64, 401.44) = (0.7994, 0.8010)$ ✓.
6. $\hat{z}_w = c_w/(\mu_w + \lambda)$, and c_w is the only thing that changes between samples, so $\Delta\hat{z}_w^{\text{ridge}} = \Delta c_w/(\mu_w + \lambda) = \frac{\mu_w}{\mu_w + \lambda}\Delta\hat{z}_w^{\text{OLS}}$. $\lambda = 1$: $0.02/1.02 = 0.0196$, i.e. ~51× smaller ($0.566 \to 0.011$).
7. $\eta = 0.04$: strong $1 - 1.6008 = -0.6008$ (|·| < 1, stable, oscillating), weak $1 - 0.0008 = 0.9992$. $\eta = 0.06$: strong $1 - 2.401 = -1.401$, |·| > 1 ⇒ **diverges** (η must be $< 2/40.02 = 0.05$).

8. $\lambda = 10$: Phase 1 ⇒ $(0.8 - 0.5, 1.2) = (0.3, 1.2)$. $\lambda = 30$: Phase 2 ⇒ $(0, (40.048 - 30)/20.04) = (0, 0.5014)$.
9. Ridge $b/2$: 0.35, 1.5. Lasso $\max(|b| - 1, 0)$: 0, 2.
10. OLS: $\hat{\omega}_1$ becomes ×10 (0.8 → 8 in new units); same predictions. Ridge: a large coefficient is penalised heavily, so ridge shifts credit toward x_2 (the cheaper coefficient), changing the fit.
11. New observations must be transformed with the **same** numbers used in training; otherwise the coefficients are applied to differently-scaled inputs.
12. Scaling with all rows → leakage, optimistic error. Choosing λ by training error → always $\lambda = 0$. Penalising the intercept → every prediction pulled toward 0. Reading the test error repeatedly → test error becomes optimistic (a second validation set).

Next: Lecture 9: back to classification. Given fitted probabilities, how do we choose a threshold and judge the decisions?